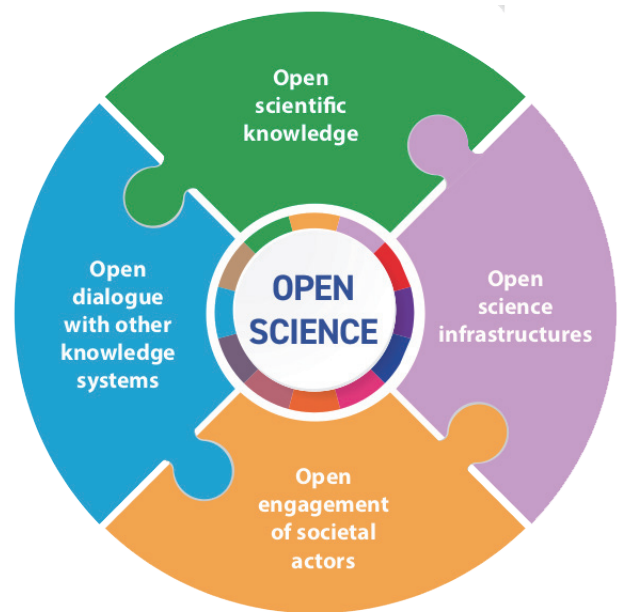


2024 ASIAN BIODIVERSITY DATA USE GUIDE

Linking to the UN 17 Sustainable Development Goals and Open Science



GBIF ASIA

GBIF CAS (Chinese Academy of Sciences) Node

March 2025



UN Sustainable Development Goals (SDGs)

In 2015, world leaders agreed to 17 goals for a better world by 2030. We recognise that eradicating poverty in all its forms and dimensions, including extreme poverty, is the greatest global challenge and an indispensable requirement for sustainable development. All countries and all stakeholders, acting in collaborative partnership, will implement this plan. We are resolved to free the human race from the tyranny of poverty and want and to heal and secure our planet. We are determined to take the bold and transformative steps which are urgently needed to shift the world onto a sustainable and resilient path. As we embark on this collective journey, we pledge that no one will be left behind. The 17 Sustainable Development Goals and 169 targets which we are announcing today demonstrate the scale and ambition of this new universal Agenda.

The official website: <https://www.globalgoals.org/>



Open science

Following an inclusive, transparent and multistakeholder consultative process, the UNESCO Recommendation on Open Science was adopted by the 41st session of UNESCO General Conference in November 2021. Open Science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

The official website: <https://www.unesco.org/en/open-science>

GBIF Data Status in Asia

There are currently 155,057,103 occurrences in Asia, 5.02 % of total GBIF records
(By 31 March, 2025)

TOP 10 Countries or Areas have most occurrences in Asia

Country or area	Count	
India	51,938,566	
Chinese Taipei	21,641,748	
China	11,239,039	
Japan	8,106,994	
Korea, Republic of	7,656,481	
Israel	7,187,525	
Thailand	6,554,989	
Russian Federation	4,666,177	
Malaysia	3,949,795	
Indonesia	3,139,006	

TOP 10 Countries or Areas contributed data in Asia

Publishing country or area	Count	
India	49,046,459	
Chinese Taipei	20,795,395	
United States of America	11,719,873	
Japan	7,454,251	
China	7,200,521	
Israel	7,199,616	
Korea, Republic of	7,021,601	
United Kingdom of Great Britain and Northern Irela...	5,631,915	
Thailand	4,961,952	
Netherlands (Kingdom of the)	2,553,364	

TOP 10 Data Publishers contributed data in Asia

Publisher	Count	
↻ Cornell Lab of Ornithology	98,118,165	
↻ iNaturalist.org	8,120,916	
↻ Taiwan Biodiversity Research Institute	4,707,381	
↻ Chinese Academy of Sciences (CAS)	4,266,741	
↻ National Museum of Nature and Science, Japan	3,971,377	
↻ National Institute of Ecology	2,360,581	
↻ MGNify	2,239,475	
↻ Israel Nature and Parks Authority	1,904,322	
↻ Naturalis Biodiversity Center	1,648,198	
↻ National Institute of Genetics, ROIS	1,476,273	

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PREFACE

The value of data lies in its use. As the largest biodiversity occurrence data portal, the Global Biodiversity Information Facility (GBIF) is dedicated “to mobilize the data, skills and technologies needed to make comprehensive biodiversity information freely available for science and decisions addressing biodiversity loss and sustainable development”.

To date, GBIF has shared over 3.09 billion records and has contributed to the production of more than 24,327 journal articles. However, despite being the largest Continent on the earth, Asia has contributed only 155,082,617 occurrences, accounting for a mere 5.02 % of the total GBIF records. Various factors have hindered the sharing of biodiversity occurrence data from this region. It is imperative to engage more individuals, institutions and organizations within the region to promote contributions to biodiversity data sharing and mobilization efforts. Urgent action is needed to address this gap and ensure broader participation in this crucial endeavor.

According to the "2023-2027 GBIF Strategic Framework", GBIF will primarily focus on four key areas: science and research, policy and partnerships, community and capacity, infrastructure and data products. Currently, two significant global initiatives, namely the UN Sustainable Development Goals (SDGs) and United Nations Educational, Scientific and Cultural Organization (UNESCO) Open Science, rely heavily on open data and other research activities. GBIF is poised to increase its contributions to these endeavors in the future. In fact, GBIF has been selected as one of the knowledge sharing platforms of UNESCO Open Science plan.

To advance the goal of supporting SDGs and Open Science, it's important to showcase various data uses and applications through a guide or handbook. Consequently, the teams of GBIF Asia and GBIF Chinese Academy of Sciences (CAS) Node have compiled 35 case studies from academic articles, categorizing them based on different SDGs, Open Science elements, and areas. I am pleased to introduce this work to all the colleagues involved in biodiversity science and conservation. I would like to take this opportunity to congratulate the authors on this achievement!

Keping MA
Head of delegation, GBIF CAS Node
Vice Chair & Secretary General, Biodiversity Committee, CAS
Professor of Institute of Botany, CAS

12 March, 2025

INTRO

The 17 UN Sustainable Development Goals (SDGs) represent the global aspiration for a better life and a more sustainable future. Open Science serves a way to engage diverse stakeholders and research endeavors in achieving these goals. Biodiversity stands as a crucial element for human beings, the planet and SDGs. The era of big data and artificial intelligence (AI) is upon us, and will significantly influence digital transformation and cooperation.

Presently, various large biodiversity data portals, such as the Global Biodiversity Information Facility (GBIF), have been established. However, there exist disparities in data sharing across regions. Notably, Asia, the largest continent, contributes only 5.02% of the total GBIF records. It's urgent to involve more individuals in data sharing efforts to ensure a sustainable future for our planet.

To provide exemplary guides, the team from National Science Library, Chinese Academy of Sciences and the Institute of Botany, Chinese Academy of Sciences have curated articles published in 2024 from diverse sources, such as Web of Science (WoS), Dimensions, and the GBIF literature database. These articles are annotated with classifications and tags, with 1-3 cases highlighted for each goal. Moreover, we ensure representation from the six sub-regions of Asia: East Asia, West Asia, Southeast Asia, Central Asia, South Asia and North Asia. Each case study incorporates elements of Open Science, including Open Scientific Publication(OSP), Open Research Data(ORD), Open Source Software (OSS), Open Educational Resources(OER) and Citizen Science(CS).

This guide is dedicated to researchers and students committed to the protection and management of biodiversity, as well as to policymakers seeking to understand the intersections between biodiversity and the SDGs/Open Science. It also extends to all citizens, professionals, and institutions, organizations joining in the collective challenge of addressing social and governance needs while safeguarding the planet.

On behalf of the Editorial Committee

Zheping Xu
Nodes Regional Representative Asia
Node Manager of GBIF CAS

12 March, 2025

HOW TO CITE: Zheping Xu, Meiling Piao, Qiqi Liu, Xin Peng, Yuning Liu, Yilin Wu(2025)2024 Asia Biodiversity Data Use Guide: Linking to the UN 17 Sustainable Development Goals and Open Science.

ACKNOWLEDGEMENTS: Base and Talent Special Project of Ministry of Science and Technology (2022XJKK1505)

1.1 TRADITIONAL KNOWLEDGE OF WILD PLANTS ON TRADITIONAL TOOLS, MATERIALS, PRODUCTS AND ECONOMIC PRACTICES IN SOUTHERN YEMEN

Yemen

Challenge

The traditional knowledge in southern Yemen is rich in wild medicinal and food plants. People in most parts of the country still depend on nature more than relying on tools and continuous materials due to war and poverty. However, the ethnobotany of this traditional knowledge is faster to be lost in the face of urbanization. In addition, other significant and general traditional usage for the daily livelihood requirements of local people (beyond medicinal and food plant uses) has not been studied before and needs urgent documentation.

Contribution

1.The ethnobotanical data resulted various traditional uses of 73 plant species distributed in 28 families in southern Yemen. The most represented families were Fabaceae, Asteraceae and Malvaceae. Nine major traditional uses of local wild plants were recorded: handicraft, agriculture tools, products aid general health, economic products, construction timber, livestock husbandry, bee keeping, fuel and ornamental. 2.Traditional knowledge of wild plants still requested vital to many usages of traditional life and still have an economic value and heritage required of develop the daily livelihood level of the local people especially in rural areas.

Citation

Al-Fatimi M. (2024). Traditional knowledge of wild plants on traditional tools, materials, products and economic practices in southern Yemen. *Journal of ethnobiology and ethnomedicine*, 20(1), 62.
<https://doi.org/10.1186/s13002-024-00698-5>



Data Uses

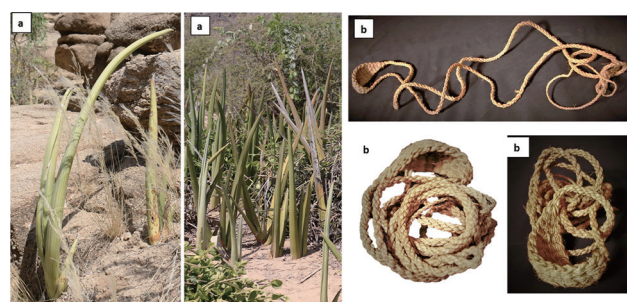
Scientific names of studied plant species and the conservation status of the reported plant species were confirmed by using GBIF (<https://www.gbif.org/>). Also used Southern Yemen flora and World Flora Online (<http://www.worldforaonline.org>) (WFO, 2021)

Open Sciences

Open scientific publications, Open research data

Affiliations

Department of Pharmacognosy, Faculty of Pharmacy, University of Aden, Yemen



a) Wild plant as a natural source for fibers used traditionally in southern Yemen. b) Traditional sling prepared traditionally from fibers of leaves of *Dracaena hanningtonii*.



1.2 ECOLOGICAL DYNAMICS, ETHNOBOTANICAL SIGNIFICANCE, AND HABITAT PROJECTIONS FOR *ARISAEMA COSTATUM* (WALL.) MART. IN RESPONSE TO CLIMATE CHANGE IN NEPAL

Nepal

Challenge

Climate change, anthropogenic pressure, deforestation and land use changes have led to continuous changes in the distribution and habitats of *Arisaema costatum*, which are ecologically and ethnobotanically very importance in the rural areas of Nepal. Therefore, it is crucial to assess critical populations and habitats to protect this species, which are under immense collection pressure, and ensure their future existence and survival.

Contribution

A. costatum plays a vital role in household food and nutrition, income generation, and health security for households in the Nepal Himalayas. Also for commercial purpose, with the final product ("Sinki" or "Gundruk") sold for at NPR 2000/kg in the local market. Furthermore, *A. costatum* serves as a vital food source for endangered and vulnerable frugivorous wild birds and porcupines in the hills and mountains of Nepal. However, elevation, annual mean temperature, and precipitation during the warmest and coldest quarters were identified as the most influential factors for projecting the future distribution of *A. costatum* in the Nepal Himalayas.

Citation

Thapa, S., Awasthi, M., Karki, S., Poudel, B. D., & Chung, K. W. (2024). Ecological dynamics, ethnobotanical significance, and habitat projections for *Arisaema costatum* (Wall.) Mart. in response to climate change in Nepal. *Global Ecology and Conservation*, 50, e2829.

<http://doi.org/10.1016/j.gecco.2024.e02829>

Data Uses

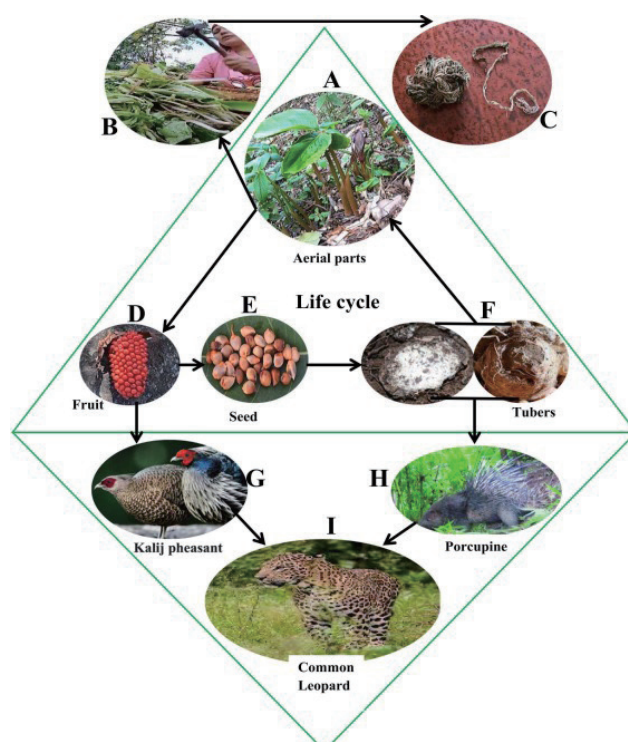
105 occurrence points from physical observations conducted across seven districts, seven records from the GBIF, <https://www.gbif.org>, accessed on 7th July, 2022, 19 from the National Herbarium and Plant Laboratories, three from Flora of Nepal (<https://floraofnepal.org/data/specimens>), and 65 points collected by the authors themselves during their duration of other fieldwork in various districts.

Open Sciences

Open scientific publications, Open research data, Open source software

Affiliations

- 1.Tri-Chandra Multiple Campus, Department of Botany, Tribhuvan University, Nepal
- 2.Department of Biological Sciences and BK21 Center for Field-oriented BioCore Human Resources Development, Kongju National University, South Korea
- 3.National Trust for Nature Conservation, Nepal
- 4.Environmental Services Nepal, Nepal



A. costatum life cycle and used by different organisms

1.3 FROM MARGINAL LANDS TO BIOFUEL BOUNTY: PREDICTING THE DISTRIBUTION OF OILSEED CROP *IDESIA POLYCARPA* IN SOUTHERN CHINA'S KARST ECOSYSTEM

China

Challenge

Biofuels are gaining attention as alternative energy sources. The Chinese government encourages planting non-food crops on marginal lands to safeguard food security and support the biofuel sector. The Southern China Karst Region, one of the most impoverished rural regions, confronting harsh natural conditions, scarcity of land, and ecological deterioration. Given the scarce and barren land, relying on traditional food crop cultivation hinders effective economic growth. Therefore, it is urgent to adopt new strategies and methods to adapt to local conditions and promote sustainable economic development.

Contribution

1. *Idesia polycarpa*, as a fast-growing tree species that is drought-tolerant and can thrive in poor soil, is well adapted to the karst region and has important value for ecological restoration, local economic development, and the advancement of China's biofuel industry.
2. The identified areas of Sichuan, Chongqing, etc. present a substantial potential for cultivation *I. polycarpa*, which surpasses that of current production zones, indicating a broader scope for the expansion of this species as a non-food energy crop.

Citation

Wu, Y., Yuan, P., Li, S., Guo, C., Yue, F., Luo, G., Yang, X., Zhang, Z., Zhang, Y., Yang, J., Wu, H., & Zhou, G. (2024). From Marginal Lands to Biofuel Bounty: Predicting the Distribution of Oilseed Crop *Idesia polycarpa* in Southern China's Karst Ecosystem. *Agronomy*, 14(7), 1563.
<https://doi.org/10.3390/agronomy14071563>

1 NO
POVERTY



Data Uses

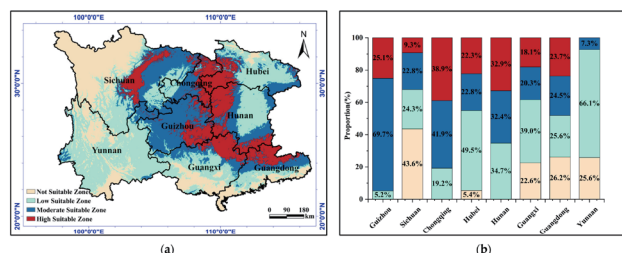
544 distribution data of the species of *I. polycarpa* were obtained from the Chinese Virtual Herbarium (CVH, <https://www.cvh.ac.cn> accessed on 1 June 2023), the China Specimen Sharing Platform (CSSP, <http://www.nsii.org.cn> accessed on 15 July 2023), and the GIBF (GBIF, <https://www.gbif.org> accessed on 16 October 2023).

Open Sciences

Open scientific publications, Open research data

Affiliations

1. School of Geography and Resources, Guizhou Education University, China
2. School of Earth system Science, Tianjin University, China
3. College of Ecology and Environment, Xinjiang University, China
4. Guizhou Provincial Key Laboratory of Geographic State Monitoring of Watershed, Guizhou Education University, China
5. Administration of Ecology and Environment of Haihe River Basin and Beihai Sea Area, Ministry of Ecology and Environment of People's Republic of China, China
6. Department of Geography and Spatial Information Techniques, Ningbo University, China
7. School of Environmental and Life Sciences, Nanning Normal University, China
8. Haihe River Water Conservancy Commission, Ministry of Water Resources of People's Republic of China, China



a) spatial distribution of suitable areas for *I. polycarpa* in the of the Southern China Karst R. b) proportion of suitable areas for *I. polycarpa* in each province of the Southern China Karst Region.

2.1 THE LESSER YAM *Dioscorea esculenta* (LOUR.) BURKILL: A NEGLECTED CROP WITH HIGH FUNCTIONAL FOOD POTENTIAL

India, Vanuatu

Challenge

In the humid tropical zone, there is a growing need for staple crops diversification and for alternatives to cereals. The lesser yam *Dioscorea esculenta* presents the highest agro-economic potential among underexploited yam species. It is now widespread throughout the wet tropics, where is concentrated the majority of the World population, and plays a role for food security in developing countries.

Contribution

D. esculenta is high yielding, robust, and offers good returns with low inputs. When plants are staked, yields of 30–40 t/ha are frequent. The small size of its tubers, and their ease of peeling, are attractive for urban consumers. It will continue to play a major role for food security and a high valued functional food in Asia and in the Pacific.

Citation

Lebot, V., Abraham, K., & Melteras, M. (2024). The lesser yam *Dioscorea esculenta* (Lour.) Burkill: a neglected crop with high functional food potential. *Genetic Resources and Crop Evolution*, 1-21. <https://doi.org/10.1007/s10722-024-02296-6>

2 ZERO HUNGER



Data Uses

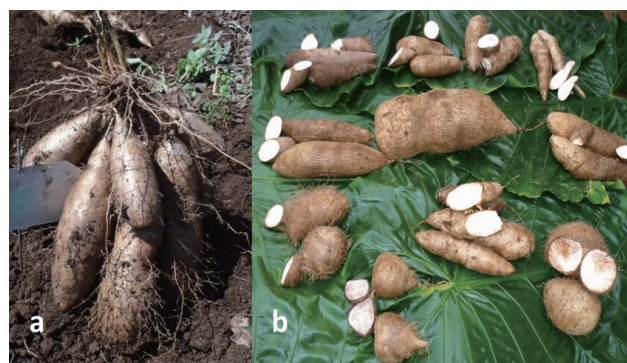
Information gained from the literature review was compared with field data recorded in CTCRI (Central Tuber Crops Research Institute), India and VARTC (Vanuatu Agricultural and Technical Research Center). GBIF 2024 was used as taxonomic clarification and geographical distribution records of *Dioscorea* species.

Open Sciences

Open research data, Open scientific publications, Open educational resources

Affiliations

- 1.CIRAD-VARTC, UMR AGAP Institute, Vanuatu
- 2.UMR AGAP Institute, University of Montpellier, France
- 3.CTCRI, Central Tuber Crops Research Institute, India
- 4.VARTC, Vanuatu Agricultural Research and Technical Center, Vanuatu



a) Selected cultivar 'Rogers' from Vanuatu with smooth tubers in tight clump easing the harvest. b) variation in tuber forms (different cultivars from Vanuatu)



3.1 ECOLOGICAL CHANGES EXACERBATING THE SPREAD OF INVASIVE TICKS HAS DRIVEN THE DISPERSAL OF SEVERE FEVER WITH THROMBOCYTOPENIA SYNDROME VIRUS THROUGHOUT SOUTHEAST ASIA

Southeast Asia

Challenge

While the tick species *Haemaphysalis longicornis* has been widely accepted as the principal vector for Severe fever with thrombocytopenia syndrome virus (SFTSV), the role of other tick species, such as *Amblyomma testudinarium*, *H. flava*, and *H. hystricis* that have also been associated with SFTSV transmission, deserve further attention. The role of mammalian hosts in SFTSV ecology is also not fully understood.

Contribution

1. This study employs a multidisciplinary approach, integrating viral phylodynamics, continuous phylogeography, and landscape phylogeographic methods to investigate the evolution and spread of SFTSV. It combines genomic data with analytical and modeling techniques to examine the interactions between SFTSV and various climate variables and ecological factors.

2. The research provides insights into the evolutionary history of SFTSV and explores the impact of natural and human-induced environmental changes on its transmission and geographic distribution. The findings help predict the potential trajectory of SFTSV spread, guide future research, and support public health interventions.

Citation

Pérez, L. J., Baele, G., Hong, S. L., Cloherty, G. A., & Berg, M. G. (2024). Ecological Changes Exacerbating the Spread of Invasive Ticks has Driven the Dispersal of Severe Fever with Thrombocytopenia Syndrome Virus Throughout Southeast Asia. *Molecular Biology and Evolution*, 41(8).
<https://doi.org/10.1093/molbev/msae173>

Data Uses

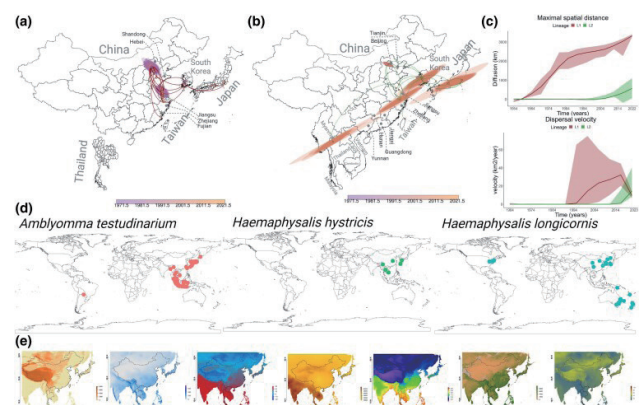
Worldpop (<https://hub.worldpop.org>), Climate Data Guide website (<https://climatedataguide.ucar.edu>), NASA POWER database (<https://power.larc.nasa.gov>), Copernicus Climate Data Store (<https://cds.climate.copernicus.eu>), NASA's Earth Observing System Data and Information System (<https://neo.gsfc.nasa.gov>), NOAA STAR Vegetation Health Product (<https://www.star.nesdis.noaa.gov>), Tick occurrence data for the species *Ixodes persulcatus* were gathered using the GBIF API with the `rgbif` package in R.

Open Sciences

Open scientific publications, Open research data, Open source software

Affiliations

1. Infectious Disease Research, Abbott Diagnostics Division, Abbott Laboratories, USA
2. Abbott Pandemic Defense Coalition (APDC), USA
3. Department of Microbiology, Immunology and Transplantation, Laboratory of Clinical and Evolutionary Virology, Rega Institute, Belgium



Tracing the historical dispersal and landscape dynamics of SFTSV lineages in Eastern Asia.



3.2 MOLECULAR INVESTIGATION OF *LEISHMANIA* IN SANDFLIES (DIPTERA: PSYCHODIDAE) AND RODENTS (MAMMALIA: RODENTIA) IN NAHAVAND, WEST OF IRAN

Thailand

Challenge

Cutaneous leishmaniasis caused by different species of *Leishmania* is transmitted by Phlebotominae sandflies. The number of patients suffering from leishmaniasis is estimated at 12 to 15 million, with 2 million new cases of leishmaniasis occurring annually. This disease remains a public health concern in Iran. The control of cutaneous leishmaniasis has been the top priority for Iran's health system for decades, but despite many efforts, not only it was not controlled but also became more prevalent with the introduction of new foci in different parts of the country.

Contribution

1. This study investigates *Leishmania* infection in sandflies and reservoir rodents in six rural regions of Nahavand, western Iran. The findings contribute to a better understanding of *Leishmania* transmission dynamics in endemic rural areas.
2. The abundance of potential *Leishmania* vectors and detection of *Leishmania major* in *P. papatasi* in the study area highlights the risk of establishment of a new CL focus in this western region considering ongoing warming and climate changes.

Citation

Zafari, S., Motavallihaghi, S., Salehzadeh, A., Zahernia, A., Sazmand, A., & Maghsood, A. H. (2024). Molecular investigation of *Leishmania* in sandflies (Diptera: Psychodidae) and rodents (Mammalia: Rodentia) in Nahavand, west of Iran. *Parasitology Research*, 123(6), 253.
<https://doi.org/10.1007/s00436-024-08265-3>

Data Uses

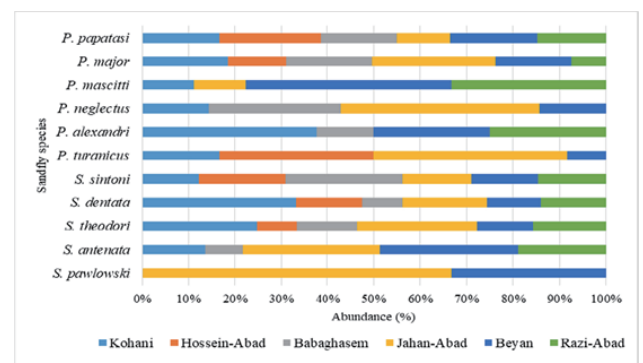
Valid names of the sandfly and rodent species were obtained from GBIF

Open Sciences

Open research data

Affiliations

1. Department of Medical Parasitology and Mycology, School of Medicine, Hamadan University of Medical Sciences, Iran
2. Department of Medical Entomology, School of Medicine, Hamadan University of Medical Sciences, Iran
3. Department of Pathobiology, Faculty of Veterinary Medicine, Bu-Ali Sina University, Iran



Abundance of sandfly species captured in Nahavand county, Hamadan province, west of Iran



4.1 CONTENT ANALYSIS OF NATURE DOCUMENTARIES IN CHINA: CHALLENGES AND OPPORTUNITIES TO RAISE PUBLIC CONSERVATION AWARENESS

China

Challenge

Rapid urbanization, however, has reduced natural areas within urban environments, leading to a steep decline in direct experiences of nature. This could potentially account for the current disaffection with the natural world and destructive behaviors, which may underlie current environmental issues. To generate public conservation efforts, the need to reconnect people with nature and raise their awareness on the ongoing biodiversity crisis has thus never been more urgent.

Contribution

1. This study compiles a comprehensive list of nature documentaries available on major streaming platforms in China. It compares the content focus of domestically produced Chinese-language documentaries with imported, predominantly English-language documentaries.
2. Assess gaps in thematic, geographic and species coverage of nature documentaries in China to maximize their contribution to conservation.

Citation

Wei, H., Berdejo-Espinola, V., Ma, Y., & Amano, T. (2024). Content analysis of nature documentaries in China: Challenges and opportunities to raise public conservation awareness. *Biological Conservation*, 292, 110522.

<https://doi.org/10.1016/j.biocon.2024.110522>

Data Uses

The scientific name identified based on the GBIF database (GBIF.org, 2022).

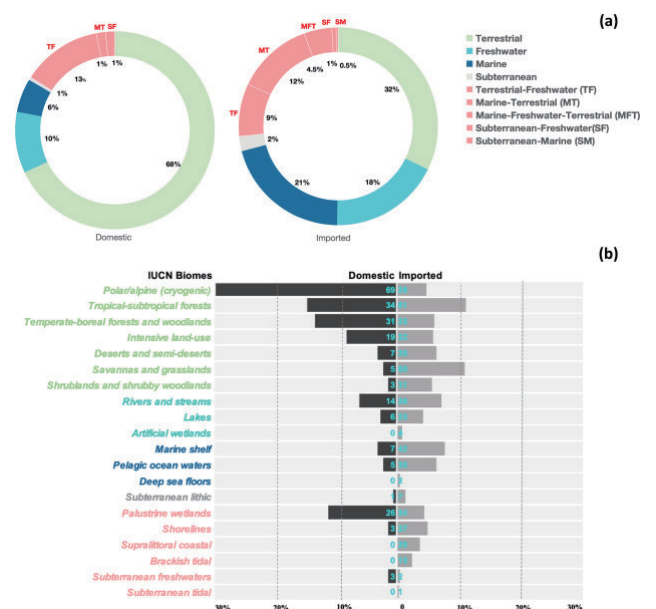
IUCN Red List of Threatened Species, Catalogue of Life (COL) , Storage platform (CCTV: <https://www.cctv.com>; BiliBili: <https://www.bilibili.com>; Youku: <https://www.youku.com>; Tencent Vedio: <https://v.qq.com>)

Open Sciences

Open scientific publications, Open research data

Affiliations

1. School of the Environment, The University of Queensland, Australia
2. Centre for Biodiversity and Conservation Science, The University of Queensland, Australia
3. Beijing University of Technology, China



a) The realm representation between domestic (left, n = 197 times of appearance in 171 documentaries) and imported nature documentaries (right, n = 289 times of appearance in 114 documentaries). b) Biome representation in domestic (dark grey bars, n = 233 times of appearance in 171 documentaries) and imported nature documentaries (light grey bars, n = 517 times of appearance in 114 documentaries).



5.1 CRITICAL ASSESSMENT OF BIOREFINERY APPROACHES FOR EFFICIENT MANAGEMENT AND RESOURCE RECOVERY FROM WATER HYACINTHS FOR SUSTAINABLE UTILIZATION

Global

Challenge

Water hyacinth is well known for seriously harming the environment and being difficult to control. Several conventional techniques for the management of Water hyacinth include physical, chemical, and biological approaches. Unfortunately, after considering the possible measures for years, these weeds have been proven difficult and expensive to control, and very little success has been achieved so far. Therefore, focusing on novel strategies to manage invasive plants still requires extensive investigation.

Contribution

1. It critically examines various technologies of used for resource recovery from water hyacinth. And its constituent elements have been employed for fabricating a broad range of commodities such as phytoremediation agents, biogas, bioethanol, biochar, biopolymers, compost, bio-fertilizers, animal feed, fish feed, high-value chemicals, and craft products.
2. It considers the impact of these technologies on social development and women's empowerment. Involvement in industrial crop processing procedures can help women become more capable of contributing substantially to home finances and decision-making, ultimately resulting in long-term empowerment.

Citation

Madhumidha, M., Benish Rose, P. M., Nagabalaji, V., Das, I., & Srinivasan, S. V. (2024). Critical assessment of biorefinery approaches for efficient management and resource recovery from water hyacinths for sustainable utilization. *Reviews in Environmental Science and Bio-Technology*, 23(2), 443-469.
<https://doi.org/10.1007/s11157-024-09693-4>

Data Uses

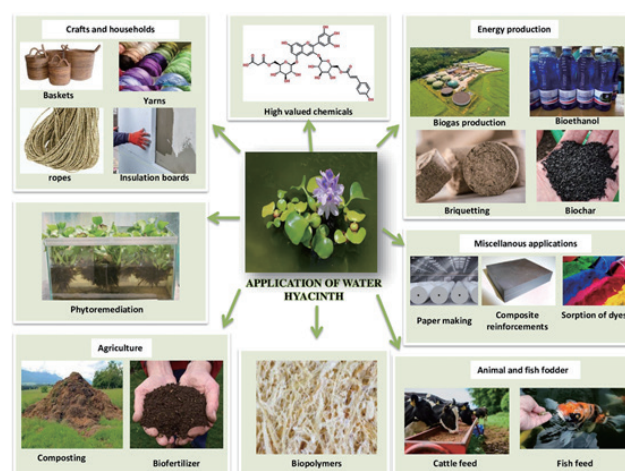
Geographical distribution of WH in the world from last 100 years (1923–2022) (from GBIF database) (Registry-Migration. Gbif. Org 2023)

Open Sciences

Open research data

Affiliations

1. Environmental Engineering Department, CSIR - Central Leather Research Institute, India
2. Academy of Scientific and Innovative Research (AcSIR), India



Possible applications of water hyacinths.



6.1 PRIORITISING CHALLENGES AND ACTIONS FOR FRESHWATER CONSERVATION IN A TROPICAL BIODIVERSITY HOTSPOT

Indonesia, Malaysia, Brunei, Singapore, Thailand

Challenge

Tropical fresh waters are experiencing one of the highest rates of biodiversity loss globally. Effective tropical freshwater biodiversity conservation requires prioritised and concerted action that is informed by science. However, efforts to synthesize the available expertise and knowledge remain lacking to date.

Contribution

This study contributes to tropical freshwater biodiversity conservation by identifying and ranking critical challenges, such as deforestation, insufficient biodiversity data, and low conservation priority, to guide targeted action. It employs the Delphi technique to build expert consensus, ensuring a reliable foundation for decision-making. Furthermore, the study proposes an integrated roadmap that combines research, stakeholder collaboration, public outreach, and policy implementation, providing a scalable model for other biodiversity hotspots facing similar threats.

Citation

Zieritz, A., Gibbins, C., Cai, Y., Diba, F., Gan, L. X., Lopes-Lima, M., Mendoza, J. C. E., Morse, J., Ng, T. H., Toh, E. X. P., Pfeiffer, J., Low, B. W., Marwoto, R., Rahim, K. A. A., Shellman, B., Sulaiman, Z., Tan, Z. W., Wowor, D., Yusuf, N. S., & Yeo, D. C. J. (2024). Prioritising challenges and actions for freshwater conservation in a tropical biodiversity hotspot. *Biological Conservation*, 299, 110839.

<https://doi.org/10.1016/j.biocon.2024.110839>

Data Uses

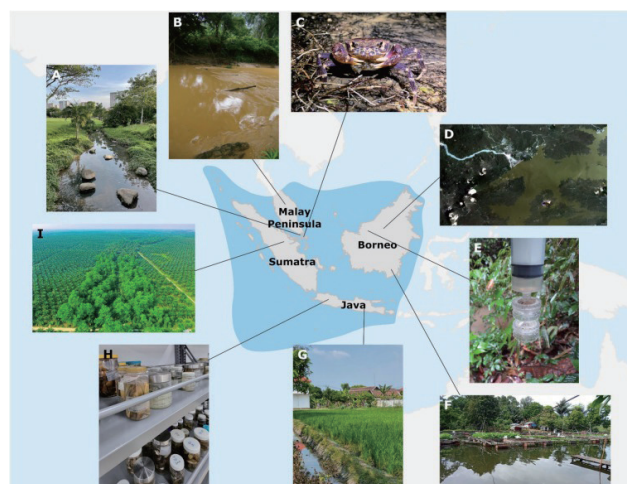
The IUCN Red List of Threatened Species (IUCN, 2022), The World Database on Protected Areas (WDPA, 2023), Global Forest Watch (2023), Malaysian Qualifications Register (2023)

Open Sciences

Open scientific publications, Open research data, Open educational resources

Affiliations

- 1.School of Geography, University of Nottingham, University Park, UK
- 2.School of Environmental and Geographical Sciences, University of Nottingham Malaysia, Malaysia
- 3.National Biodiversity Centre, National Parks Board, Republic of Singapore
- 4.Tanjungpura University, Faculty of Forestry, Indonesia
- 5.Lee Kong Chian Natural History Museum, Faculty of Science, National University of Singapore, Republic of Singapore
- 6.BIOPOLIS Program in Genomics, Biodiversity and Ecosystems, CIBIO, Centro de Investigação em Biodiversidade e Recursos Genéticos, InBIO Laboratório Associado, Campus de Vairão, Universidade do Porto, Portugal



Examples of challenges for freshwater biodiversity conservation across the biodiversity hotspot Sundaland and/or potential solutions to address them.



6.2 UNVEILING THE URBAN COLONIZATION OF THE ASIAN WATER MONITOR (*VARANUS SALVATOR*) ACROSS ITS DISTRIBUTION RANGE USING CITIZEN SCIENCE

Sri Lanka, Indonesia, Thailand, Bangladesh

Challenge

Urbanization, a major driver of habitat transformation, presents significant challenges by creating new ecological pressures on wildlife, influencing dispersal patterns, reproductive success, dietary habits, predation risks, and parasite-host interactions. However, research remains heavily skewed toward cities in Europe, North America, and Australia, with a strong focus on plants, birds, and mammals, while reptiles, particularly in tropical and developing regions, receive far less attention.

Contribution

This study highlights the understudied presence of the Asian water monitor (*Varanus salvator*) in urban environments across its distribution range, revealing its preference for green patches like parks and gardens within cities. Despite being the largest lizard with established urban populations, research on the species remains limited, primarily focusing on diet and behavior. The study underscores the role of urban design in shaping its distribution and emphasizes the need for further research to understand both ecological and human-related dimensions of its urban adaptation. Additionally, it demonstrates the value of citizen science data in identifying species presence and distribution patterns in cities.

Citation

Luna, Á., Rausell-Moreno, A. (2024). Unveiling the urban colonization of the Asian water monitor (*Varanus salvator*) across its distribution range using citizen science. PeerJ, 12, e17357. <http://doi.org/10.7717/peerj.17357>

Data Uses

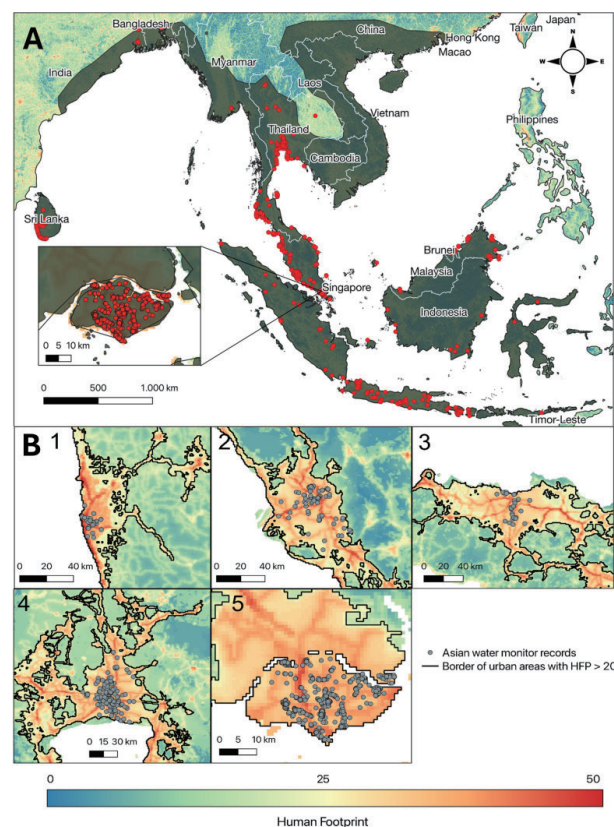
Occurrence data of Asian water monitors from the GBIF (<https://doi.org/10.15468/dl.sg6x5g>, <https://doi.org/10.15468/dl.78mgfn>), iNaturalist (<https://www.inaturalist.org/>), Google Earth Engine (<https://earthengine.google.com/>), Moderate Resolution Imaging Spectroradiometer (MODIS) (<https://modis.gsfc.nasa.gov/>), eBird (<https://ebird.org/>)

Open Sciences

Open scientific publications, Open research data, Open educational resources

Affiliations

1. Department of Biosciences, Universidad Europea de Madrid, Spain
2. Department of Biogeography and Global Change, National Museum of Natural Sciences (MNCN-CSIC), Spain



Records of urban monitor lizards (*V. salvator*) in its native range. a) Distribution range of the Asian water monitor (*V. salvator*) in grey.



7.1 INVESTIGATING THE SUSTAINABLE ENERGY GENERATION POTENTIAL OF AN INVASIVE WEED: *LANTANA CAMARA*

India

Challenge

Lantana camara, one of the world's top ten most invasive species, was initially cultivated for ornamental use. However, it spread uncontrollably across the fallow areas and agricultural lands, threatening approximately 44% of Indian forests. Its invasion disrupts ecosystems by suppressing nearby plant growth through allelopathy and poses toxicity risks to grazing ruminants. It significantly increases forest fire risk by adding large amounts of combustible biomass, particularly dried *L. camara*. Despite efforts to control it using mechanical, chemical, and biocontrol methods, the results have been largely unsatisfactory, with associated costs estimated at \$18,000 per square kilometre.

Contribution

This study demonstrated that using *L. camara* for bioenergy production presents a dual solution, addressing the growing demand for renewable energy and managing invasive species. This review aims to critically evaluate the potential and challenges associated with the different energy production pathways for *L. camara*, highlighting its role in sustainable energy generation.

Citation

Kaushik, Y., Arora, P. (2024). Investigating the sustainable energy generation potential of an invasive weed: *Lantana camara*. *Environmental Science and Pollution Research*, 31(54), 62493–62509. <https://doi.org/10.1007/s11356-024-35322-2>

Data Uses

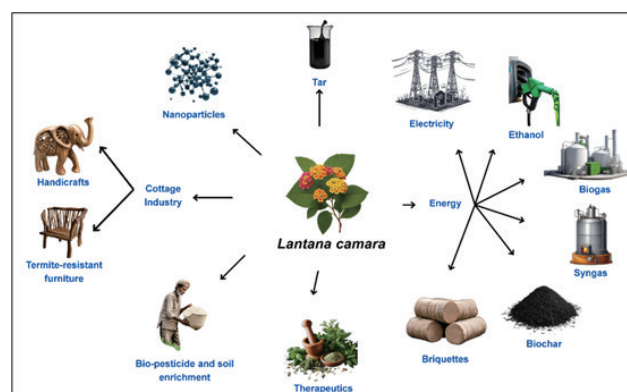
GBIF (<https://doi.org/10.15468/dl.pbf7f>), Web of Science (WOS) (<https://webofscience.clarivate.cn/>)

Open Sciences

Open scientific publications, Open educational resources

Affiliations

Hydro and Renewable Energy Department, Indian Institute of Technology Roorkee, India



Various applications of *L. camara*



7.2 BIOCLIMATIC ANALYSIS OF POTENTIAL WORLDWIDE PRODUCTION OF SPRING-TYPE CAMELINA [*CAMELINA SATIVA* (L.) CRANTZ] SEEDED IN THE SPRING

Global

Challenge

Camelina (*Camelina sativa* L. Crantz) is a Brassicaceae oilseed that is gaining interest worldwide as low-maintenance crop for diverse biobased applications. In particular, camelina has been identified as a feasible option for sourcing a low indirect Land Use Change feedstock for sustainable aviation fuel. One of the most important factors determining its productivity is climate. It's imperative to gain an understanding of the variability and adaptation potential of camelina across environments.

Contribution

This study developed a bioclimatic model using CLIMEX to analyze the productivity of camelina in relation to climatic factors. Sensitivity analysis highlights that camelina's yield potential is more influenced by water availability than temperature, emphasizing its suitability for water-limited conditions. The study identifies new and expanded production regions, such as China, eastern Russia, offering valuable insights for global camelina cultivation, supporting its further potential as a sustainable crop and fuel for biobased applications.

Citation

Weiss, R. M., Zanetti, F., Alberghini, B., Puttick, D., Vankosky, M. A., Monti, A., Eynck, C. (2024). Bioclimatic analysis of potential worldwide production of spring-type camelina [*Camelina sativa* (L.) Crantz] seeded in the spring. *Global Change Biology Bioenergy*, 16(2), e13126.
<https://doi.org/10.1111/gcbb.13126>

Data Uses

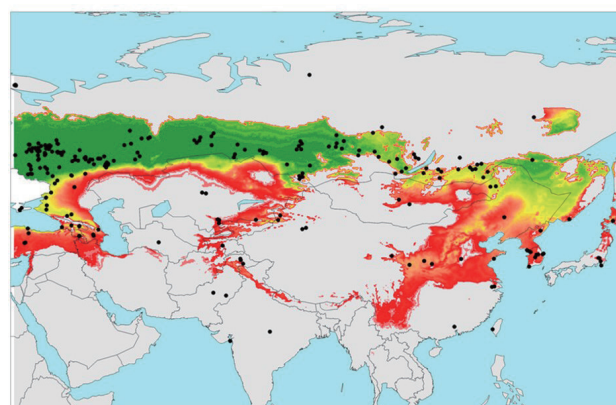
GBIF, Crop protection compendium (CABI) (<https://www.cabi.org/>), Germplasm Resources Information Network (GRIN) (<https://www.ars-grin.gov/>)

Open Sciences

Open scientific publications, Open research data, Open educational resources

Affiliations

1. Agriculture and Agri-Food Canada - Saskatoon Research and Development Center, Canada
2. Department of Agricultural and Food Sciences, Alma Mater Studiorum, University of Bologna, Italy
3. Smart Earth Camelina Corporation, Canada



Distribution data, projected distribution and ecoclimatic index results for Asia



8.1 ECOLOGY, ECONOMIC BOTANY AND CONSERVATION OF *DIPLOKNEMA BUTYRACEA* IN NEPAL

Nepal

Challenge

Diploknema butyracea is a multipurpose cultural keystone tree species (CKTS). In Nepal, *D. butyracea* tree and its different parts have been utilized and conserved by ethnic groups of Nepal for centuries. Besides cultural accounts, the *D. butyracea* also has an important role in biodiversity, recreational values, and the household economy of the Chepang community. About one-third of the households of Raksirang have maintained their household income with annual average of US\$ 2000 by selling *D. butyracea* products. But, the current degradation and deforestation, coupled with overexploitation and incursion of alien invasive plants have a significant effect on the conservation and restoration of *D. butyracea*.

Contribution

This study systematically reviews over 102 research studies and conducts extensive field visits to document the distribution, ethnobotanical uses, and ecological significance of *D. butyracea*, particularly among the Chepang community. By collaborating with government officials, forest user groups, and local communities across 62 districts, the study provides a comprehensive foundation for conservation and highlights the need for integrating traditional knowledge with scientific research to promote biodiversity and bioprospecting.

Citation

Bhattarai, S., Bhatta, B., Shrestha, A. K., Kunwar, R. M. (2024). Ecology, economic botany and conservation of *Diploknema butyracea* in Nepal. *Global Ecology and Conservation*, 51, e2869.
<https://doi.org/10.1016/j.gecco.2024.e02869>

Data Uses

Georeferences data collected through National Herbarium & Plant Laboratories (KATH), Royal Botanic Gardens, Kew (K), Natural History Museum (NHM), Kyoto Herbarium (KYO), GBIF and iNaturalist.

Open Sciences

Open scientific publications, Open research data, Open educational resources, Citizen science

Affiliations

1. Faculty of Forestry, Agriculture and Forestry University, Nepal
2. Faculty of Agriculture, Agriculture and Forestry University, Nepal
3. Gandaki University, Nepal



Plant habit and inflorescence of *Diploknema butyracea*.



9.1 RESERVOIRS ALTER TERRESTRIAL MAMMAL HABITAT OVER THE INDOCHINA PENINSULA

China, India

Challenge

Dams and their reservoirs play a critical role in social and economic development as they can supply seasonal water needs or generate renewable energy. The global prosperity of reservoirs has resulted in widespread habitat loss and degradation, but little attention has been given to the survival crisis of terrestrial biota. Also, the systematic and explicit impacts of reservoir construction on terrestrial mammals in this area are largely unknown.

Contribution

This study provides a comprehensive analysis of the impacts of reservoir development on terrestrial mammalian habitats. It reveals that reservoirs significantly alter habitat structure and function, increasing fragmentation and reducing connectivity. The identification of 29 priority obstacle points for removal offers actionable insights to mitigate these impacts. By emphasizing the need to balance reservoir development with biodiversity conservation, the study calls for policy-makers and energy strategists to consider ecological impacts in key biodiversity hotspots, particularly in the Indochina Peninsula, to prevent further habitat destruction and biodiversity loss.

Citation

Lan, X., Zhou, T., Zeng, T., Chen, Z., Duo, J., & Sun, J. (2024). Reservoirs alter terrestrial mammal habitat over the Indochina Peninsula. *Ecological Indicators*. 166, 112366.

<https://doi.org/10.1016/j.ecolind.2024.112366>

Data Uses

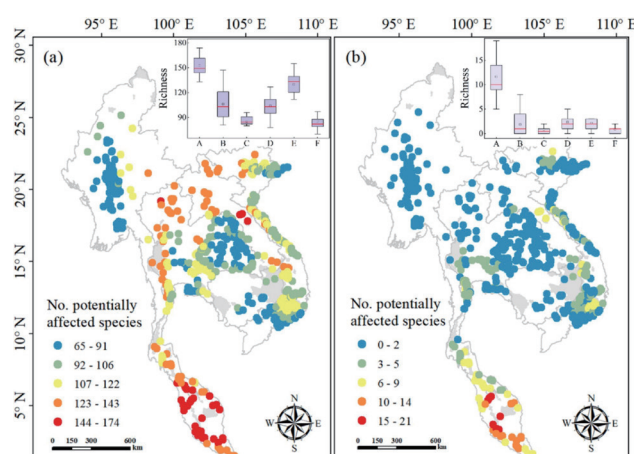
GBIF, Global Reservoir and Dam dataset (GOODD), MODIS Land Cover Type Product (MCD12Q1) Version 6, WorldClim Global Climate Data (WorldClim), World Database on Protected Areas (WDPA)

Open Sciences

Open scientific publications, Open research data, Open educational resources

Affiliations

- 1.State Key Laboratory of Tibetan Plateau Earth System, Environment and Resources, Institute of Tibetan Plateau Research, Chinese Academy of Sciences, China
- 2.College of Ecology, Lanzhou University, China
- 3.Center for Pan-third Pole Environment, Lanzhou University, China
- 4.School of Earth and Environmental Sciences, China
- 5.College of Earth Sciences, Chengdu University of Technology, China
- 6.School of Life Sciences, Qinghai Normal University, China
- 7.Haixi Forest Pest Control and Quarantine Center, China



Distribution of existing reservoirs across the mammal distribution in the Indochina Peninsula.



9.2 OPTIMIZATION OF MULTIPLE ECOLOGICAL INFRASTRUCTURES ACROSS THE LAND–SEA INTERFACE FOR COORDINATION MANAGEMENT: A CASE STUDY AROUND LAIZHOU BAY IN CHINA

China

Challenge

Ecological infrastructure (EI), providing ecosystem services across the land-sea interface, has been proposed as a key element in sustainable terrestrial-marine ecosystem coordinated governance. Terrestrial and marine ecosystems should be regarded as an integrated unit for guaranteeing coastal ecological security. However, the existing EI construction framework focused on terrestrial ecosystems, and few studies consider the composite characteristics of the terrestrial-marine ecosystem in coastal areas.

Contribution

1. This study contributes to sustainable coastal ecosystem management by proposing an innovative optimization framework for multiple ecological infrastructures (MEIs) across the land-sea interface, using Laizhou Bay, China, as a case study.
2. The framework integrates terrestrial multi-scale ecological networks, hydrological corridors, and marine conservation networks, enhancing connectivity between terrestrial, riverine, and marine ecosystems.
3. The study also pinpoints conserved priority areas, including ecological pinch points and marine biological protective points, providing actionable spatial references for coordinated terrestrial-marine ecosystem management. This approach offers a feasible and comprehensive solution for promoting ecological material circulation and ensuring coastal ecological security.

Citation

Qiu, G., Wang, J., Liu, J., & Wang, X. (2024). Optimization of multiple ecological infrastructures across the land-sea interface for coordination management: A case study around Laizhou Bay in China. *Science of the Total Environment*. 949, 175105. <https://doi.org/10.1016/j.scitotenv.2024.175105>

Data Uses

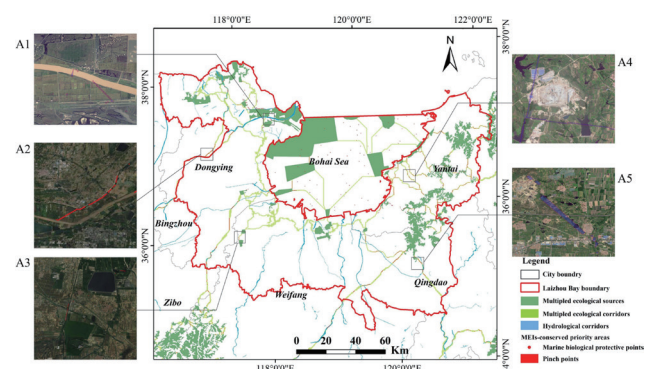
Data type	Format	Data source
Land use	Tiff	Resources and Environment Data Center, Chinese Academy of Sciences
NPP	Tiff	MOD17A3 product from NASA-USGS
Soil	Tiff	Harmonized World Soil Database v 1.2
Meteorological	Tiff	China National Meteorological Information Center (CNMIC)
DEM	Tiff	National Centers for Environmental Information
River system	Shp	OpenStreetMap
Marine functional zoning	Shp	Shandong Province Marine Functional Zone Planning (2011–2020)
Seawater quality	Shp	Shandong Provincial Department of Ecology and Environment
Marine data products	Tiff	NASA, National Aeronautics and Space Administration, agency of US government
Remote sensing	Tiff	Geospatial Data Cloud
Species distribution	Shp	Global Biodiversity Information Platform GBIF

Open Sciences

Open scientific publications, Open research data

Affiliations

- 1.College of Water Sciences, Beijing Normal University, China
- 2.Key Research Institute of Yellow River Civilization and Sustainable Development, Henan University, China



Distribution and location of MEIs-conserved priority areas around Laizhou Bay

10.1 A LATITUDINAL GRADIENT OF REFERENCE GENOMES

Global

Challenge

Unequal global scientific resource allocation results in severe genomic data deficits in low-latitude biodiversity hotspots. Technological barriers in genomics exacerbate methodological disparities in molecular ecology research between the Global North and South. The lack of high-quality reference genomes for tropical endangered species hinders the application of conservation genetics.

Contribution

1. First quantification reveals significant discrepancies between the distribution of tetrapod reference genomes and latitudinal gradients of species diversity.
2. Low-coverage whole-genome sequencing applications in Global South research account for only 1.5%.
3. Proposes feasibility plans for establishing tropical species genome resource repositories through transnational collaborative projects.

Citation

Linck, E. B., & Cadena, C. D. (2024). A Latitudinal Gradient of Reference Genomes. *Molecular ecology*, e17551.
<https://doi.org/10.1111/mec.17551>



Data Uses

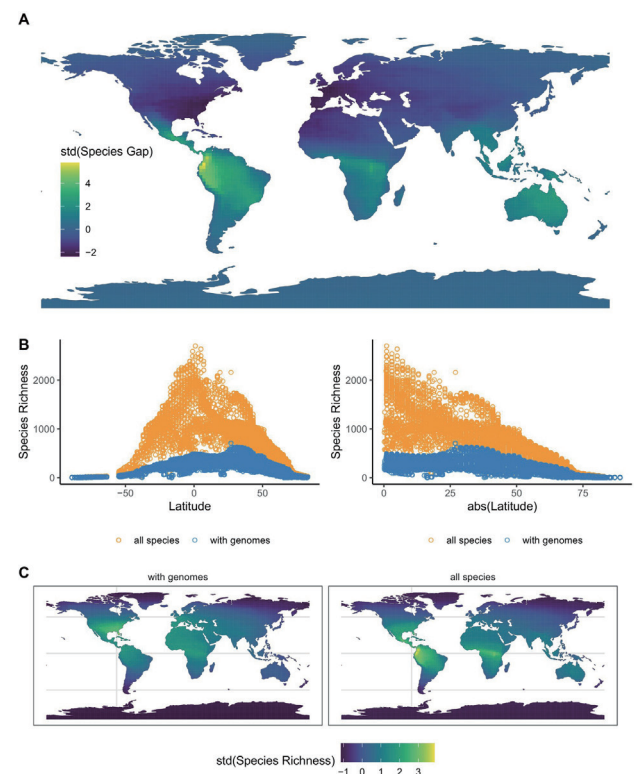
GBIF Backbone Taxonomy using rgbif v.3.8.0 (Chamberlain et al. 2024), <https://doi.org/10.15468/dl.p4r6hf>, <https://doi.org/10.15468/dl.njrdnn>, <https://doi.org/10.15468/dl.zw99uq>, <https://doi.org/10.15468/dl.jbg6xy>, <https://doi.org/10.15468/dl.5pkrsd>, <https://doi.org/10.15468/dl.jd85b2>, <https://doi.org/10.15468/dl.59eyey>, <https://doi.org/10.15468/dl.vybgce>. National Center for Biotechnology Information (NCBI) Datasets command-line tools v.16.19.0

Open Sciences

Open scientific publications, Open research data, Open source software

Affiliations

1. Department of Ecology, Montana State University, USA
2. Department of Biology, Universidad de los Andes, Colombia



a) Tropical species are underrepresented among available reference genomes. b) Richness among species with reference genomes does not reflect the global patterns of biodiversity.

11.1 TROPICAL CITIES AS WINDOWS INTO THE ECOSYSTEMS OF OUR PRESENT AND FUTURE

Global



Challenge

Global urbanization research exhibits geographical bias, with severe gaps in ecological data from tropical cities. Tropical species are more sensitive to urban heat island effects and habitat fragmentation, yet species-specific traits and community response mechanisms remain poorly characterized. Green spaces in tropical cities are unevenly distributed, with affluent neighborhoods often monopolizing ecological resources. Cities simultaneously function as refuges for native species and hubs for invasive species proliferation.

Contribution

1. Proposes the "Tropical Urban Ecological Resilience" theory, emphasizing the adaptability of indigenous species. Redefines "nature–culture" boundaries by incorporating cultural landscapes such as temple forests and graveyards into conservation frameworks. Advocates for "eco-inclusive planning" by mandating 20% retention of native vegetation in new urban developments.
2. Develops a hybrid urban green volume estimation model integrating optical and radar data to address remote sensing limitations caused by tropical cloudy weather. Proposes an "Urban Wildlife Corridor Certification System" to promote the protection of transboundary migratory species
3. Validates urban biodiversity hotspots using community science data (e.g., citizen-led bird monitoring initiatives).

Citation

Bonebrake, T. C., Tsang, T. P. N., Yu, N., Wang, Y., Ledger, M. J., Tilley, H. B., Yau, E. Y. H., Andersson, A. A., Boyle, M. J. W., Lee, K. W. K., Li, Q., Ling, Y. F., Dongmo, M. A. K., Güçlü, C., Dingle, C., & Ashton, L. A. (2025). Tropical cities as windows into the ecosystems of our present and future. *Biotropica*, 57(1) <http://doi.org/10.1111/btp.13369>

Data Uses

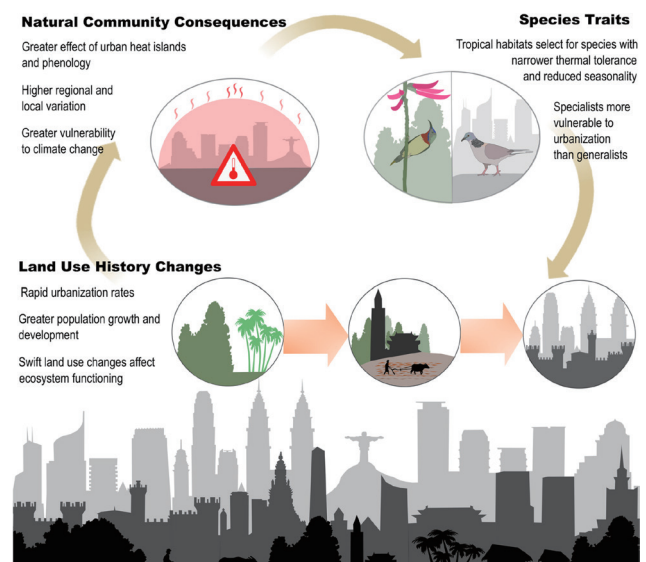
GBIF (<https://doi.org/10.15468/ab3s5x>), iNaturalist, Burungnesia and Kupunesia, eBird, European Commission Joint Research Centre (JRC), Norway's International Climate and Forest Initiative (NICFI) Satellite Data, Resource Watch, Google EarthEngine.

Open Sciences

Open scientific publications, Open research data

Affiliations

1. School of Biological Sciences, The University of Hong Kong, China
2. Department of Biological Sciences, University of Toronto-Scarborough, Canada
3. Urban Big Data Centre, University of Glasgow, UK
4. School of Humanities and Social Science, The Chinese University of Hong Kong, China



Testable hypotheses related to tropical urban species.



11.2 ON THE HOME-VISITING DRAGONS: A YEAR-ROUND OBSERVATION OF DRAGONFLIES (INSECTA: ODONATA) ENTERING AN URBAN RESIDENCE

Indonesia

Challenge

Urbanization processes have led to the encroachment of natural habitats, forcing dragonflies and other insects into urban environments in search of survival resources. Artificial nighttime lighting exacerbates light pollution, disrupting the circadian activity rhythms of dragonflies and particularly affecting crepuscular activity patterns and migratory behaviors of nocturnal species. Methodological limitations include the study's reliance on a single urban residential site and a relatively short temporal scale, resulting in a small sample size, which may constrain the generalizability of conclusions.

Contribution

1. Documented the first year-round record of dragonfly intrusion in Padang City, identifying 12 species, including rare taxa, and revealing peak activity during evenings and nights linked to artificial lighting.
2. Highlighted an equal sex distribution in intrusions, challenging dispersal behavior assumptions, and advocated for sustainable urban planning to mitigate human-wildlife conflicts, emphasizing dragonflies as bioindicators of environmental health.

Citation

Janra, M. N., Herwina, H., & Jefrial. (2024). On the home-visiting dragons: a year-round observation of dragonflies (Insecta: Odonata) entering an urban residence. *IOP conference series. Earth and environmental science*, 1306(1), 12006
<http://doi.org/10.1088/1755-1315/1306/1/012006>

Data Uses

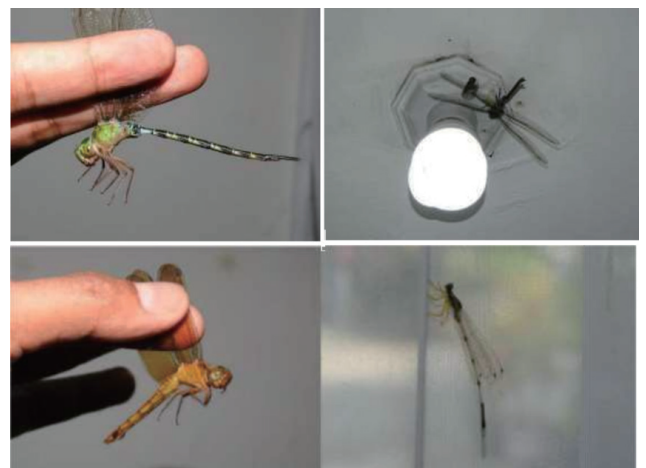
Orthetrum sabina Drury, 1773 GBIF secretariat 2021 GBIF backbone taxonomy *Checklist Dataset*, <https://doi.org/10.15468/39omei> available: GBIF.org on 2021-07-05. Czech: Taita Publisher, Yogyakarta: Indonesia Dragonfly Society.

Open Sciences

Open scientific publications, Open research data

Affiliations

Biology Department, Faculty of Mathematics and Natural Sciences, Universitas Andalas, Indonesia



Documentation of intruded Odonata: male *Gynacantha dorrhni* (top left), the same individual perched next to a light bulb (top right), female *Rhodothemis* sp (bottom left), and male *Copera marginipes* (bottom right).



11.3 ETHNOBOTANY OF WILD EDIBLE PLANTS USED BY LOCAL COMMUNITIES IN THREE DISTRICTS ALONG THE UPPER BENGAWAN SOLO RIVER, CENTRAL JAVA, INDONESIA

Indonesia

Challenge

Urban expansion encroaches upon natural habitats, jeopardizing the survival of wild edible plant resources. Intergenerational erosion of traditional knowledge threatens the continuity of orally transmitted indigenous wisdom, as younger generations increasingly disconnect from ancestral practices. Globalization-driven commercialization of food systems diminishes reliance on wild plant resources, undermining cultural and nutritional resilience. Methodologically, geographically confined sampling and a modest sample size may limit broader applicability. Furthermore, reliance on subjective respondent accounts, despite cross-verification efforts, risks potential biases or incomplete data capture.

Contribution

1. Highlights mounting tensions between ecological preservation and economic development, demanding urgent exploration of pathways for sustainable wild plant utilization amidst escalating land-use pressures and environmental degradation. Reveals the holistic integration of WEPs into daily life, with plants serving as sources of carbohydrates, vegetables, fruits, seeds, and medicine.
2. Links plant use to socio-cultural dynamics, such as generational knowledge gaps and gender roles. Emphasizes the threats posed by urbanization, including habitat loss and pollution, alongside the need for sustainable land-use planning to preserve biodiversity and cultural heritage.

Citation

TRIYANTO, A., PURNAMASARI, F., PARAMITA, F. S., WICAKSONO, F. R., RAMADHAN, F. A., BUDIHARTA, S., SAENSOUK, S., & SETYAWAN, A. D. (2024). Ethnobotany of wild edible plants used by local communities in three districts along the upper Bengawan Solo River, Central Java, Indonesia. *Biodiversitas Journal of Biological Diversity*, 25(4) <https://doi.org/10.13057/biodiv/d250428>

Data Uses

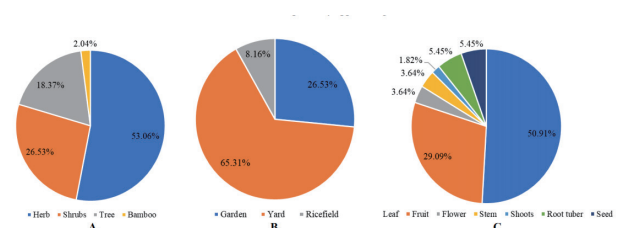
Scientific name identified by using GBIF and Powo. Data was also collected by conducting open, semi-structured, and structured interviews directly by visiting key and ordinary informants' homes.

Open Sciences

Open scientific publications, Open research data

Affiliations

1. Department of Biology, Faculty of Mathematics and Natural Sciences, Universitas Sebelas Maret, Indonesia
2. Department of Environmental Science, Faculty of Mathematics and Natural Sciences, Universitas Sebelas Maret, Indonesia
3. Purwodadi Botanic Gardens, Research Center for Ecology and Ethnobiology, National Research and Innovation Agency, Indonesia
4. Plant and Invertebrate Taxonomy and Its Applications Unit Group, Biodiversity Program, Walai Rukhavej Botanical Research Institute, Mahasarakham University, Thailand
5. Biodiversity Research Group, Universitas Sebelas Maret, Indonesia



a) The percentage of plant growth forms. b) The percentage of plant habitat. c) The percentage of plant parts used in the research area.



12.1 ANALYSIS OF GLOBAL TRADE PATTERNS AND INDUSTRIAL LAYOUT IN MEDICINAL PLANTS: A CASE STUDY OF *EUCOMMIA ULMOIDES* OLIVE

Global

Challenge

China, holding 99% of global *Eucommia* resources, faces habitat loss due to climate change and overexploitation, while trade deficits arise from export dominance but rising processed goods imports. Developed nations control premium markets via high-value products, whereas Egypt struggles with fragmented supply chains. Static ecological models and lack of a dedicated HS code hinder accurate analysis. Industrial clustering in southern China and overreliance on northwest ports create logistics mismatches with EU/NA markets, necessitating diversified corridors and standardized certifications.

Contribution

1. Conducted the first systematic assessment of global medicinal plant trade structures, identifying China, India, and Egypt as key exporting nations, while the U.S., Germany, and France emerged as core import markets.
2. China's high-suitability zones faced a 66.9% shrinkage, threatening domestic production. Identified untapped cultivation potential in Belt and Road countries through climate resilience mapping, enabling strategic resource allocation and diversification. Developing northern/western China's medicinal markets to alleviate southern hubs' congestion. Aligning domestic regulations with international certifications to boost global market access.

Citation

Luo H, Zhang H, Zhang X, Zhang G, Huang L. (2025). Analysis of global trade patterns and industrial layout in medicinal plants: A case study of *Eucommia ulmoides* oliver. *Industrial Crops and Products*, 225, 120555. <http://doi.org/10.1016/j.indcrop.2025.120555>

Data Uses

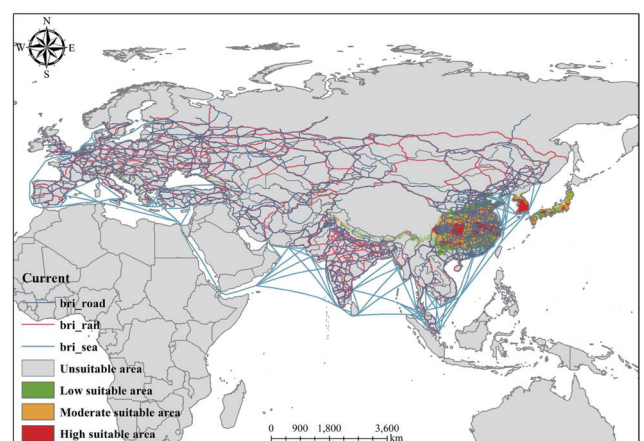
China Chamber of Commerce for Import & Export of Medicines & Health Products (CCCMHPIE), the United Nations Commodity Trade Database, China's trade data of *E. ulmoides* commodities is collected from the CCCMHPIE, the baseline distribution data obtained from the GBIF, the Chinese Virtual Herbarium (CVH) and National Specimen Information Infrastructure (NSII)

Open Sciences

Open scientific publications, Open research data, Open source software

Affiliations

1. Key Laboratory of Chinese Medicine Resources Conservation, State Administration of Traditional Chinese Medicine of the People's Republic of China, Institute of Medicinal Plant Development, Chinese Academy of Medical Sciences & Peking Union Medical College, China
2. Jiangxi University of Chinese Medicine, China



Transportation routes of *E. ulmoides* along the Belt and Road countries in the current period.

12.2 SPECIES DIVERSITY AND CONSERVATION STATUS OF ORNAMENTAL FISH TRADED IN KUPANG, EAST NUSA TENGGARA, INDONESIA

Indonesia

Challenge

The ornamental fish trade in Kupang, Indonesia, faces significant challenges including data gaps in conservation status for 8 species, risks from invasive non-native species, unsustainable harvesting of endangered taxa like *Scleropages formosus* and *Sahyadria denisonii*, weak enforcement of trade regulations, limited public awareness, and socioeconomic pressures prioritizing economic gains over ecological sustainability. These issues are compounded by taxonomic uncertainties, small sample sizes in surveys, and the absence of systematic monitoring for released species, necessitating urgent interdisciplinary solutions to balance biodiversity conservation with livelihoods.

Contribution

1. Highlights Kupang City as a critical hub for ornamental fish trade in East Nusa Tenggara. Emphasizing the first systematic documentation of 46 species, including hybrid and endangered taxa. This comprehensive approach underscores the city's significant role in the regional and potentially global ornamental fish market.

2. Underscores the vulnerability of 5 species and the need for targeted protection measures, and implicitly addresses the dual role of trade in economic growth versus ecological risk. This analysis provides a foundational baseline for sustainable management, offering actionable insights for policymakers and stakeholders to balance economic benefits with ecological sustainability.

Citation

USMAN, Z., HARIYADI, D. R., & SERIHOLLO, L. G. G. (2024). Species diversity and conservation status of ornamental fish traded in Kupang, East Nusa Tenggara, Indonesia. *Biodiversitas Journal of Biological Diversity*, 25(3), 1116-1126
<https://doi.org/10.13057/biodiv/d250326>



Data Uses

Catalogue of Life, 2023, Encyclopedia of Life (EOL), 2023; GBIF; IUCN; CITES

Open Sciences

Open scientific publications, Open research data

Affiliations

Species diversity and conservation status of ornamental fish traded in Kupang, Indonesia



Ornamental fish traded in Kupang City. 1. *Scleropages formosus*, 2. *Osteoglossum bicirrhosum*, 3. *Amphilophus citrinellus* x *Cichlasoma trimaculatum*, 4. *Astronotus ocellatus*, 5. *Pterophyllum scalare*, 6. *Betta splendens*, 7. *Sahyadria denisonii*, 8. *Polypterus Endlicheri*, 9. *Epalzeorhynchos frenatus*, 10. *Boraras maculatus*, 11. *Pangasianodon hypophthalmus* (Albino), 12. *Moenkhausia sanctaefilomenae*, 13. *Pseudotropheus Socolofi*, 14. *Carassius auratus* (Komet), 15. *Chindongo demasoni*, 16. *Danio rerio* (Glofish)



13.1 HABITAT LOSS IN THE IUCN EXTENT: CLIMATE CHANGE-INDUCED THREAT ON THE RED GORAL (*NAEMORHEDUS BAILEYI*) IN THE TEMPERATE MOUNTAINS OF SOUTH ASIA

Myanmar, India, China

Challenge

Climate change poses significant threats to the Red Goral, including rapid habitat loss and severe fragmentation, while only 22.29% of suitable habitats are within protected areas—some of which, like Mouling NP, may vanish entirely under future scenarios. Political instability in Myanmar undermines conservation efforts, highlighting the need for international cooperation to address illegal wildlife trade and habitat destruction. Human activities, such as hydropower projects in India causing habitat loss and traditional hunting/medicinal trade, exacerbate these risks. Additionally, limited data in remote regions like Tibet reduces the accuracy of predictive models, complicating conservation planning.

Contribution

1. Identifies current suitable habitat for Red Goral at 13.01% of its IUCN range, projecting a 34–45% decline by 2080 due to climate change, with precipitation seasonality, elevation, and temperate forests as key factors. Highlights Dibang Wildlife Sanctuary and Hkakaborazi National Park as critical, resilient habitats. 2. Calls for enhanced cooperation between India, Myanmar, and China to protect shared Red Goral habitats and maintain key corridors, emphasizing the importance of transboundary conservation efforts.

Citation

Abedin I, Mukherjee T, Abedin J, Kim H, Kundu S. (2024). Habitat Loss in the IUCN Extent: Climate Change-Induced Threat on the Red Goral (*Naemorhedus baileyi*) in the Temperate Mountains of South Asia. *Biology*, 13(9), 667
<https://doi.org/10.3390/biology13090667>

Data Uses

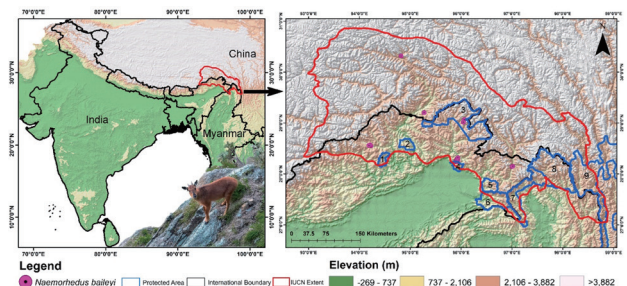
WorldClim website, Diva-GIS website, GBIF and iNaturalist

Open Sciences

Open scientific publications, Open research data; Open source software, Citizen science

Affiliations

1. Agricultural and Ecological Research Unit, Indian Statistical Institute, India;
2. Dibru-Saikhowa Conservation Society, India;
3. Department of Marine Biology, Pukyong National University, Republic of Korea
4. Marine Integrated Biomedical Technology Center, National Key Research Institutes in Universities, Pukyong National University, Republic of Korea
5. Department of Biology, Faculty of Science and Technology, Airlangga University, Surabaya 60115, Indonesia Institute of Fisheries Science, College of Fisheries Science, Pukyong National University, Republic of Korea
6. International Graduate Program of Fisheries Science, Pukyong National University, Republic of Korea



Map showing the study area for the present study along with the IUCN extent of Red Goral (*N. baileyi*).



13.2 HOW CAN DRY TROPICAL FORESTS RESPOND TO CLIMATE CHANGE? PREDICTIONS FOR KEY NON-TIMBER FOREST PRODUCT SPECIES SHOW DIFFERENT TRENDS IN INDIA

India

Challenge

Due to reliance on bioclimatic variables alone. Projections under RCP 8.5 assume high emissions but are contingent on uncertain socioeconomic and political trajectories, while regional climate variability may be underrepresented by using a single GCM. Spatially clustered occurrence data in accessible regions like Maharashtra and Madhya Pradesh risk skewing results for remote areas such as northeast India. Ecological complexities like species interactions and disturbance regimes are not accounted for, potentially affecting model accuracy. Implementing conservation strategies requires overcoming socioeconomic barriers through collaboration with local stakeholders. Additionally, limited mid-Holocene fossil pollen records constrain the comprehensiveness of paleoecological validation.

Contribution

1. Under the future climate change, the suitable habitat for *A. marmelos* and *T. bellirica* is expected to increase while for *B. lanzan*, *M. longifolia* and *P. emblica*, it is projected to decline. *A. marmelos* and *T. bellirica* are anticipated to exhibit resilience to future climate changes and are expected to be minimally affected, while *B. lanzan*, *M. longifolia* and *P. emblica* are highly sensitive to high temperature and alteration in rainfall pattern expected under future climate changes.
2. The projections of habitat suitability areas can be used as a valuable foundation for developing conservation and restoration strategies aimed at alleviating the climate change impacts on Non-Timber Forest Products species.

Citation

Saraf, P. N., Srivastava, J., Munoz, F., Charles, B., & Samal, P. (2024). How can dry tropical forests respond to climate change? Predictions for key Non-Timber Forest Product species show different trends in India. *Environmental Monitoring and Assessment*, 196(8), 727.
<https://doi.org/10.1007/s10661-024-12876-9>

Data Uses

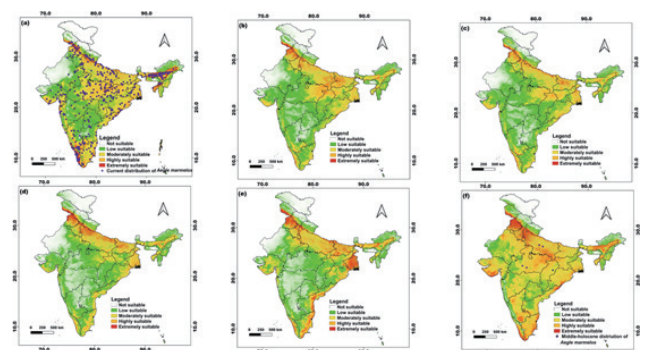
iNaturalist (www.inaturalist.org accessed 22 May 2023), Indian Biodiversity Portal (indiabiodiversity.org/ procured 2 July 2023), Indian Virtual Herbarium, GBIF Occurrence Download <https://doi.org/10.15468/dl.t2ja6c>, <https://doi.org/10.15468/dl.5ezxjs>, <https://doi.org/10.15468/dl.ntygqw>, <https://doi.org/10.15468/dl.sswq4n>, <https://doi.org/10.15468/dl.sfp4db>.

Open Sciences

Open scientific publications, Open research data, Citizen science

Affiliations

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2. Academy of Scientific and Innovative Research (AcSIR), India
3. Interdisciplinaire de Physique (LIPhy), Université Grenoble Alpes, France
4. Institute for Biodiversity Conservation and Training, India
5. CSIR-National Botanical Research Institute, India



a) The current distribution of *A. marmelos* in India. b) Predicted future distribution model of *A. marmelos* under RCP 2.6 (2050), c) RCP 2.6 (2070), d) RCP 8.5 (2050) and e. RCP 8.5 (2070), f) mid-Holocene distribution of *A. marmelos*



13.3 EFFECTS OF CLIMATE CHANGE ON THE DISTRIBUTION OF *SCOMBER JAPONICUS* AND *KONOSIRUS PUNCTATUS* IN CHINA'S COASTAL AND ADJACENT WATERS

China

Challenge

Climate change has increasingly impacted the marine environment, with various marine environmental factors interacting to influence fish distribution. Assessing the impact of climate change on the future distribution of fish depends on understanding how biological responses interact with environmental conditions. Enhancing our understanding of the potential impacts of climate change is crucial for the sustainable development of marine fisheries.

Contribution

1.The Maximum Entropy Model was used to predict the distribution of fish habitats.it can accurately predict the future distribution of suitable habitats for *Scomber japonicus* and *Konosirus punctatus* under different climate change scenarios.

2.Climate change will significantly affect the future habitat distribution of the two fish species. It is helpful for fishery managers to develop sustainable development strategies. By understanding the impact of climate change on fish distribution, managers can make advance plans.

Citation

Xia, M., Jia, H., Wang, Y., & Zhang, H. (2024). Effects of Climate Change on the Distribution of *Scomber japonicus* and *Konosirus punctatus* in China's Coastal and Adjacent Waters. *Fishes*, 9(10), 395. <https://doi.org/10.3390/fishes9100395>

Data Uses

GBIF, the Ocean Biogeographic Information System (OBIS), Fishbase, and Fishnet2

SPSS software, the Bio-ORACLE database

Open Sciences

Open scientific publications, Open research data

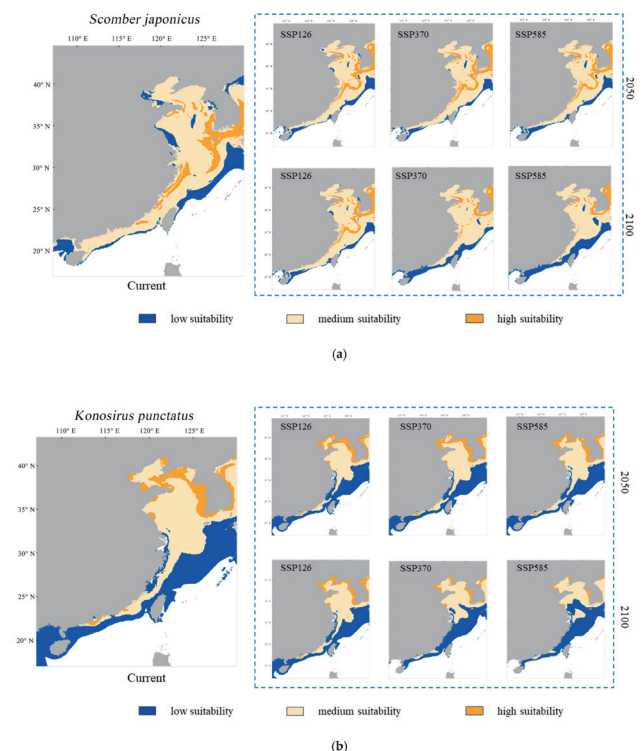
Affiliations

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3.Yangtze Estuary Marine Ecosystem Research Station, Institute of Oceanology, Chinese Academy of Sciences, China

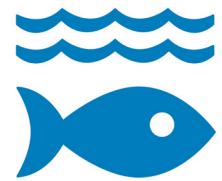
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Distribution of suitable areas for *S. japonicus* and *K. punctatus* under different climatic scenarios.

14.1 EXPLORING SPECIES AND FUNCTIONAL DIVERSITY OF FISHES IN CAMBODIAN COASTAL HABITATS USING EDNA METABARCODING

Cambodia



Challenge

Cambodian coastal ecosystems support millions of people through fisheries, tourism, shipping, and resource extraction. In coastal ecosystems such as coral reefs, seagrass beds, and mangrove forests, understanding fish biodiversity and functional diversity is essential to inform ecosystem conservation and sustainable resource management. However, traditional methods for monitoring fish biodiversity, such as direct capture and visual observation, are invasive, time-consuming, costly, and often lack the sensitivity to detect rare or elusive species, with detections being affected by weather and water conditions.

Contribution

1. This study identifies diverse fish species and underscores eDNA metabarcoding as an effective, non-invasive tool for biodiversity assessment and conservation planning.
2. The research highlights Koh Kong's high biodiversity and the presence of endangered species, supporting the establishment of marine protected areas for conservation.

Citation

Clay, C. G., Gösler, F., Glue, M., Long, M., Rog, S. M., Sinovas, P., & Reimer, J. D. (2025). Exploring species and functional diversity of fishes in Cambodian coastal habitats using eDNA metabarcoding. *Coral Reefs*, 44(1), 221-241.
<https://doi.org/10.1007/s00338-024-02599-1>

Data Uses

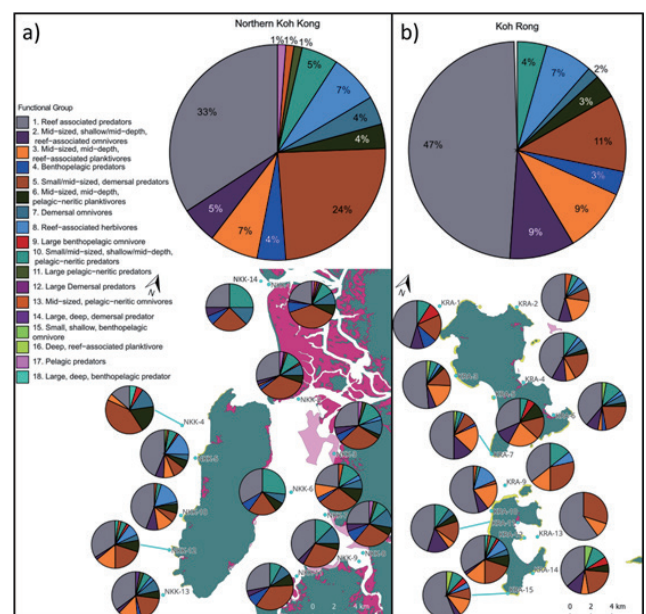
GBIF, NCBI nucleotide database, IUCN Red List, World Register of Marine Species

Open Sciences

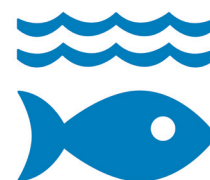
Open scientific publications, Open research data, Open source software

Affiliations

1. School of Biology, Faculty of Biological Sciences, University of Leeds, UK
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3. Fauna & Flora, Cambodia, Cambodia
4. Tropical Biosphere Research Center, University of the Ryukyus, Japan



Percentage of number of unique, detected fish species belonging to a certain functional group per site. Functional groups based on the average linkage hierarchical clustering of fish species based on five traits: maximum length, maximum depth, tropic level and habitat. Colors indicate different functional groups. a) Northern Koh Kong region. b) Koh Rong Archipelago



14.2 THE DISTRIBUTION OF THE RED-THROATED ASCIDIAN *HERDMANIA MOMUS* SHIFTS NORTHWARDS IN ASSOCIATION WITH OCEAN WARMING IN THE KOREAN PENINSULA

Republic of Korea

Challenge

Marine biodiversity in temperate seas faces several threats, including climate change and invasive species. The increasing global temperatures associated with climate change have led to notable shifts in the distribution ranges of marine species, often pushing them toward higher latitudes, unlike terrestrial organisms, marine species tend to exhibit more pronounced geographic expansion in response to climate change. Such transformations in marine ecosystems can result in complex adverse interactions.

Contribution

1. The poleward expansion of subtropical marine species into the Korean Peninsula due to climate change, at least in the case of *H. momus*, is occurring faster than previously predicted.
2. While the IPCC's RCP scenarios are widely used to predict future marine ecosystem changes, they struggle to account for localized regions experiencing rapid temperature increases, such as the Korean Peninsula.

Citation

Hwang, C., Lee, S. J., Seok, H. J., Kim, H., Hwang, I., Kang, M. G., & Park, J. M. (2024). The distribution of the red-throated ascidian *Herdmania momus* shifts northwards in association with ocean warming in the Korean Peninsula. *Journal of the Marine Biological Association of the United Kingdom* 104, e44, 1–6.
<https://doi.org/10.1017/S0025315424000365>

Data Uses

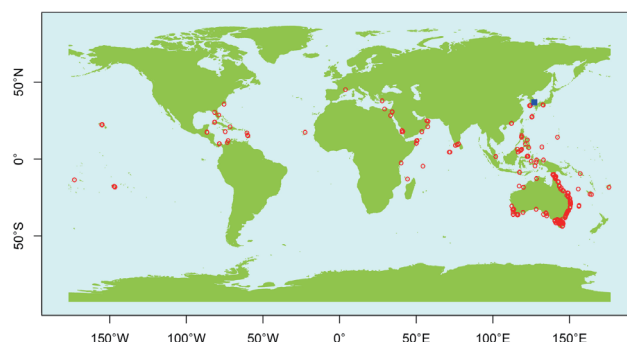
GBIF (accessed 11 September 2023), Ascidiacea World Database (<https://www.marinespecies.org/ascidiacea>), World Register of Marine Species (<https://www.marinespecies.org>)

Open Sciences

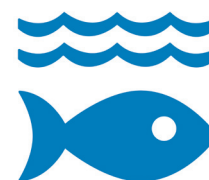
Open scientific publications, Open research data

Affiliations

1. Marine Eco-Technology Institute, Republic of Korea;
2. Marine Ecosystem Management Department, Korea Marine Environment Management Corporation, Republic of Korea;
3. Dokdo Research Center, East Sea Research Institute, Korea Institute of Ocean Science & Technology, Republic of Korea



Distributional records of *H. momus*, showing historical records (red circles) from the Global Biodiversity Information Facility (GBIF, 2023) and the new record (blue square) off southeastern coast, South Korea.



14.3 CHECKLIST OF MARINE FISHES IN THE BEIBU GULF: FISH CLASSIFICATION, RESOURCE PROTECTION, AND BIODIVERSITY CHALLENGE

China

Challenge

The Beibu Gulf, a semi-enclosed bay in the northwest of the South China Sea, bordered by China and Vietnam, features numerous harbors, islands, and is influenced by several rivers, creating a diverse ecological landscape. Despite the importance of a systems perspective for understanding resilience in marine resource use and biodiversity conservation, there has been no comprehensive survey of fish species diversity in the gulf. Consequently, data on species diversity is fragmented, incomplete, and lacks a comprehensive catalog.

Contribution

This study significantly enhances our knowledge of the Beibu Gulf's marine biodiversity, documenting 1,059 fish species and revealing 57 with uncertain distributions and eight alien species, underscoring the ecosystem's complexity and dynamism. It also highlights taxonomic challenges, with many species experiencing classification changes, and identifies research gaps, particularly along the unexplored western coasts of the Beibu Gulf and Hainan Island.

Citation

Luo, Z., Yi, M., Yang, X., Chen, X., Wang, J., Jiang, C., Liu, F., Luo, K., He, X., Lin, H., Kang, B., & Yan, Y. (2025). Checklist of marine fishes in the Beibu Gulf: fish classification, resource protection, and biodiversity challenge. *Journal of Oceanology and Limnology*, 43(1), 232-247.

<https://doi.org/10.1007/s00343-024-3210-1>

Data Uses

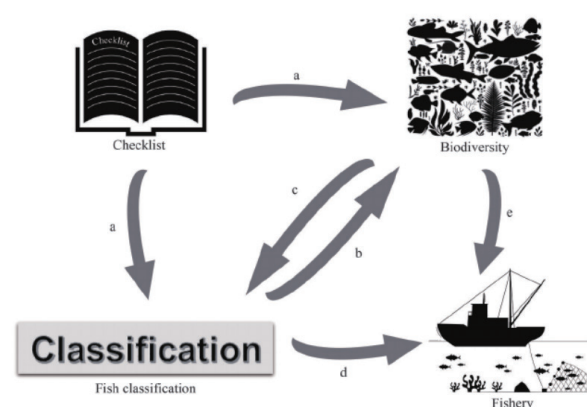
Fish species distribution data from the GBIF, Ocean Biogeographic Information System databases (OBIS, <http://www.iobis.org>), FishBase (<https://www.fishbase.se/search.php>) and IUCN Red List

Open Sciences

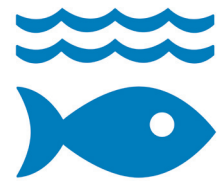
Open scientific publications, Open research data

Affiliations

1. College of Fisheries, Guangdong Ocean University, Zhanjiang 524088, China
2. College of Marine Sciences, South China Agricultural University, Guangzhou 510642, China
3. Fisheries College, Ocean University of China, Qingdao 266003, China
4. Guangdong Provincial Engineering and Technology Research Center of Far Sea Fisheries Management and Fishing of South China Sea, Guangdong Ocean University, Zhanjiang 524088, China



A checklist provides information reflecting the connection and interaction between fish classification, biodiversity, and fisheries resources



14.4 SEASONAL DISTRIBUTION PATTERNS AND CONSERVATION GAPS OF BLUE SHARKS IN THE INDO-WESTERN PACIFIC OCEAN

Asia

Challenge

Oceanic sharks, as key top predators, are vital for marine biodiversity and ecosystem stability, but human activities, particularly overfishing, have led to a 71% decline in 18 well-studied populations over the past 50 years, with three-quarters now threatened with extinction, posing serious ecological risks including trophic cascades and fishery collapses.

Contribution

1. This study finds some prominent seasonal variations in the distribution maps, suggesting that previous non-seasonal SDM studies might have masked important habitat-use dynamics over the year.
2. This study provides maps to do a dynamic approach to MPA networks for protecting oceanic sharks with seasonal habitats, and also apply bivariate hotspot analyses to identify shark-fishery conflict hotspots and conservation opportunities, in contrast to previous studies that rely on an exposure risk index

Citation

Zhang, Y., Lian, P., & Zhang, X. (2024). Seasonal distribution patterns and conservation gaps of blue sharks in the Indo-Western Pacific Ocean. *Diversity and Distributions*, 30(5), e13828. <https://doi.org/10.1111/ddi.13828>.

Data Uses

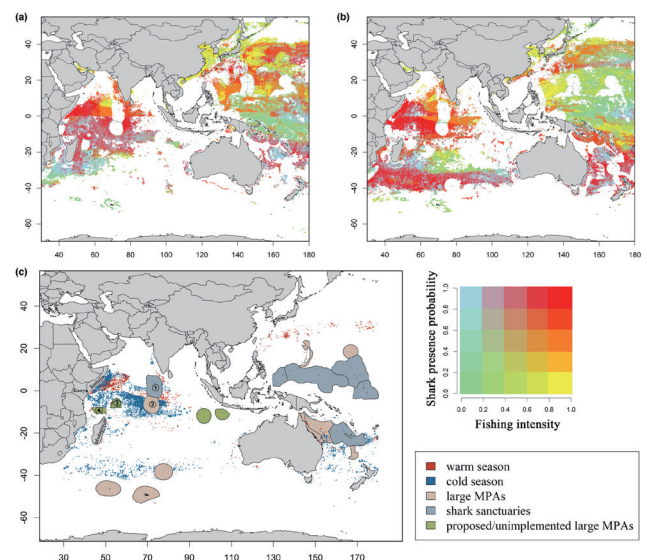
GBIF, OBIS, Copernicus Marine Service (<https://marine.copernicus.eu>), General Bathymetric Chart of the Oceans (<https://download.gebco.net>), IUCN red list of threatened species.

Open Sciences

Open scientific publications, Open research data, Open source software

Affiliations

1. School of Ecology, Sun Yat-sen University, China
2. Key Laboratory of Ocean Circulation and Waves, Institute of Oceanology, Chinese Academy of Sciences, China
3. School of Marine Sciences, Sun Yat-Sen University & Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), China



Bivariate maps of fishing intensity and blue shark presence probability in a) the warm season and b) the cold season. Colour scales are based on quantile intervals (in 20% increments) with each class containing an equal number of values. c) Twenty per cent hotspots for shark presence probability and fishing intensity in each season, overlaid with protected areas. Annotated protected areas: 1, Maldives Shark Sanctuary; 2, British Indian Ocean Territory MPA; 3, Amirantes to Fortune Bank Area of Outstanding Natural Beauty; 4, Aldabra Group Marine National Park.



15.1 CITIZEN SCIENCE FOR INVERTEBRATE BIODIVERSITY: MOBILIZING SPIDER OCCURRENCE DATA THROUGH FACEBOOK IN INDIA

India

Challenge

Spiders, a species-rich group, exemplify this knowledge gap due to the need for taxonomic revisions. Several studies on spider systematics and biogeography have highlighted the lack of data on their spatial distribution, seasonality, and natural history in India.

Contribution

1. This project showcases the capacity of citizen science via social media to involve citizen scientists in generating extensive datasets that make a significant contribution to scientific knowledge and improve our comprehension of invertebrate biodiversity.
2. The final dataset encompasses over 15,000 observations, providing valuable insights into spider diversity and distribution across India. This data is publicly available on GBIF, facilitating further research on Indian spider populations.

Citation

Barve V, Vattakaven T, Barman N, Jadhav Kulkarni N, Basu Roy A, Singh P, Kulkarni S. (2024). Citizen Science for Invertebrate Biodiversity: Mobilizing Spider Occurrence Data through Facebook in India. *Biodiversity Information Science and Standards*, 8, e139170.
<https://doi.org/10.3897/biss.8.139170>

Data Uses

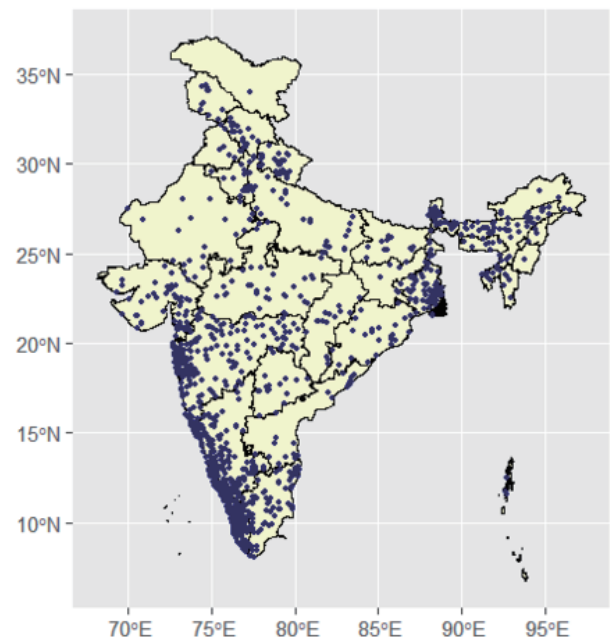
India Biodiversity Portal, Facebook

Open Sciences

Open scientific publications, Citizen science

Affiliations

1. Natural History Museum of Los Angeles County, United States of America
2. Nature Mates Nature Club, India
3. India Biodiversity Portal; Strand Life Sciences, India
4. Indian Institute of Science, Education and Research, India



Geographical distribution of occurrence records of spiders derived from Spider India Facebook group



15.2 PRIORITISING IMPORTANT PLANT AREAS (IPAS) AMONG THE LIMESTONE KARSTS OF PERAK, MALAYSIA

Malaysia

Challenge

Most limestone karsts in Perak are on state land and remain unprotected. While some sites fall under UNESCO or National Heritage protection, the majority are vulnerable to land-use changes. Even those designated as geoparks lack strict legal enforcement. Quarrying, which has been ongoing since the 1970s, continues to expand, posing significant threats to the unique biodiversity of these areas. Additionally, 59 karsts in Perak have yet to be surveyed botanically, and many rare species have not been recorded in over 30 years. This knowledge gap makes it difficult to fully assess conservation priorities. Many endemic and critically endangered species are restricted to single karsts, making them highly susceptible to extinction due to habitat destruction or environmental changes.

Contribution

1. This study systematically assessed 95 limestone karsts in Perak, identifying 17 as priority conservation areas based on a quantitative scoring system that considers endemic and threatened species, capturing over 90% of plant diversity, endemism, and threatened species, thereby providing a reliable framework for prioritizing karsts for legal protection and guiding targeted conservation actions.
2. The research highlights significant knowledge gaps in botanical surveys and species distributions, emphasizing the need for ongoing research and fieldwork to update conservation assessments, refine protection strategies, and ensure efforts are directed towards the most vulnerable plants with restricted distributions.

Citation

Tan, J. P. C., Kiew, R., & Darbyshire, I. (2024). Prioritising Important Plant Areas (IPAs) among the limestone karsts of Perak, Malaysia. *Kew Bulletin*, 79(2), 409-427. <https://doi.org/10.1007/s12225-023-10160-6>

Data Uses

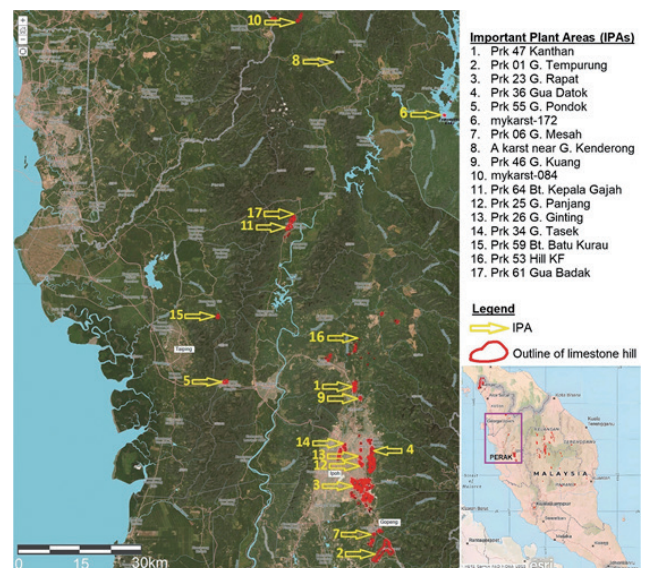
GBIF Secretariat (2016). GBIF Backbone Taxonomy. Checklist Dataset. <https://doi.org/10.15468/39omei> via GBIF.org [Accessed 3 June 2017], The IUCN Red List of Threatened Species (<https://doi.org/10.2305/IUCN.UK.1998.RLTS.T31370A962.9697.en>), Plants of the World Online (<http://www.plantsoftheworldonline.org/>), The Plant List (<http://www.theplantlist.org/>)

Open Sciences

Open scientific publications, Open research data, Open source software and source code

Affiliations

1. Kajang, selangor, Malaysia
2. Forest Research Institute Malaysia, Malaysia
3. Royal Botanic Gardens, Kew, UK



Map of 17 karsts prioritised as IPAs based on species richness and conservation priority scores



16.1 1090. *IRIS NUSAIRIENSIS* MOUTERDE: IRIDACEAE

Syria

Challenge

The IUCN Red List assessment classified *Iris nusairiensis* as Critically Endangered due to a sharp population decline, with key threats including grazing and human gathering. However, data on the species remain limited. The ongoing Syrian conflict has exacerbated habitat loss by reducing protection measures, damaging infrastructure, and increasing uncontrolled access to protected areas. Fires caused by shelling, increased demand for firewood due to power shortages, and intensified grazing—both from domestic and feral herbivores—have further degraded its habitat. At specific localities, insect herbivory has also hindered flower maturity. Additionally, military activity in the Nusayriyah range, including the construction of trenches and deployment of heavy machinery, has further disrupted the species' habitat.

Contribution

1. The study discovered three new populations of *Iris nusairiensis* in the southern Slunfe area and reassessed its extinction risk based on population size, distribution, life history, and ecological threats. This update will contribute to revising the IUCN Red List assessment.
2. The authors propose targeted conservation strategies, including strengthening ranger presence, securing funding, engaging local communities, and restoring damaged habitats to ensure the species' survival.
3. The research highlights how the Syrian conflict has led to habitat destruction, reduced protection measures, and increased environmental pressures such as overgrazing and deforestation.

Citation

Daiyoub, A., Freeth, T., Langhorne, J., Nilson, J., Hall, T. & Mas, S.S. (2024). 1090. *Iris nusairiensis* Mouterde: Iridaceae. *Curtis's Botanical Magazine*, 41(1), 41–58. <https://doi.org/10.1111/curt.12562>

Data Uses

GBIF Secreteriat (2023). *Iris nusairiensis* Mouterde checklist dataset [dataset]. GBIF Backbone Taxonomy (<https://doi.org/10.15468/39omei>), World Flora Online (<http://www.worldfloraonline.org/>), Plants of the World Online (<http://www.plantsoftheworldonline.org/>), IUCN Red List (<https://doi.org/10.2305/IUCN.UK.2016-.RLTS.T13161767A18612495.en>.)

Open Sciences

Open scientific publications, Open research data

Affiliations

1. Centre for Ecological Research and Forestry Applications (CREAF), Spain
2. Royal Botanic Gardens, Kew, Richmond, UK
3. Gothenburg Botanic Garden, Sweden



Iris nusairiensis in habitat.



16.2 CAUGHT IN THE CROSSFIRE: BIODIVERSITY CONSERVATION PARADOX OF SOCIOPOLITICAL CONFLICT

Philippines

Challenge

Conflict-affected regions often lack comprehensive biodiversity assessments, limiting the ability to establish conservation priorities. Scientists face challenges in accessing these areas due to safety concerns, further exacerbating data deficiencies. Military activities, including deforestation, infrastructure development, and resource exploitation, significantly alter ecosystems. Historical examples include the deforestation during the Vietnam War and the destruction of forests in the Democratic Republic of the Congo. Conflict leads to increased reliance on natural resources, promoting overharvesting, illegal logging, and poaching. Displaced communities and armed groups may exploit biodiversity-rich areas for survival, further threatening endangered species.

Contribution

1. The study provides quantitative analysis showing the negative correlation between conflict intensity and species richness. This evidence underscores the urgent need for conflict-sensitive conservation strategies
2. The research highlights critical biodiversity knowledge shortfalls in the Philippines, particularly in Mindanao. It emphasizes the need for better documentation, monitoring, and targeted conservation efforts in high-risk areas
3. The paper suggests 10 conservation strategies for conflict areas, including using environmental DNA, conservation drones, and bioacoustics to monitor biodiversity in inaccessible areas. These technologies offer safer alternatives for data collection in conflict-affected regions.

Citation

Hilario-Husain, B. A., Tanalgo, K. C., Guerrero, S. J. C., Garcia, F. G. N., Leros, T. E., Garcia, M. E. Z., ... & Agduma, A. R. (2024). Caught in the crossfire: biodiversity conservation paradox of sociopolitical conflict. *Npj Biodiversity*, 3(1), 10.
<https://doi.org/10.1038/s44185-024-00044-8>

Data Uses

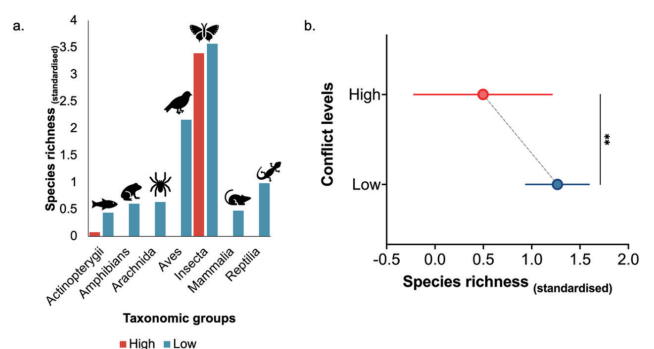
The MOBIO+: a FAIR (Findable, Accessible, Interoperable and Reusable) database for Mindanao's terrestrial biodiversity

Open Sciences

Open scientific publications, Open research data, Open source software

Affiliations

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3. Department of Mathematics and Statistics, College of Science and Mathematics, University of Southern Mindanao, Philippines.
4. Department of Social Science and Philosophy, College of Arts and Social Sciences, University of Southern Mindanao, Philippines.
5. Guangxi Key Laboratory of Forest Ecology and Conservation, College of Forestry, Guangxi University, China.
6. State Key Laboratory for Conservation and Utilization of Subtropical Agrobioresources, Guangxi University, China.



Differences in species richness between low- and high-conflict areas. a) Comparison of normalized species richness among taxonomic groups and b) overall group comparison between low- and high-conflict areas. Note: ** indicates significance at $p < 0.001$; whiskers represent 95% CI intervals.



16.3 IDENTIFYING IMPORTANT HORNBILL LANDSCAPES IN SARAWAK, MALAYSIA

Malaysia

Challenge

Sarawak and Sabah, the two Malaysian states in Borneo, host vast intact forests that support 8 of Malaysia's 10 hornbill species. However, conservation in Sarawak is urgently needed due to significant forest loss in recent decades and ongoing threats from planned developments. Additionally, knowledge gaps remain regarding key hornbill priority areas. Despite the government's commitment to curbing illegal logging, Sarawak's reliance on forestry and logging continues to pose a threat to hornbill habitats.

Contribution

1. Identified eight Important Hornbill Landscapes in Sarawak, with five being high-priority areas, providing a foundation for targeted conservation efforts.
2. Identified breeding and nest record limitations as key factors affecting IHL identification, emphasizing the need for further data collection.
3. Suggested that the eight Important Hornbill Landscapes identification framework could be applied in Sabah and other Asian regions to expand hornbill conservation prioritization, particularly in Southeast Asian countries facing similar forestry-conservation challenges.

Citation

Wee, S. Q., Teo, J. J., Teepol, B., Jelembei, H. N., Au, N. J., Yeap, C. A., & Jain, A. (2024). Identifying important hornbill landscapes in Sarawak, Malaysia. *Global Ecology and Conservation*, 50, e02828. <https://doi.org/10.1016/j.gecco.2024.e02828>

Data Uses

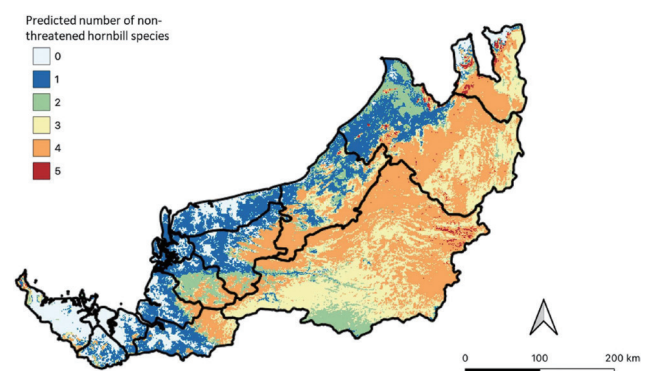
GBIF (<https://doi.org/10.15468/dl.u9etnm> & <https://doi.org/10.15468/dl.uhuy7d>), Important Bird and Biodiversity Areas list of Malaysia (<https://datazone.birdlife.org/country/malaysia/ibas>), Xeno-canto (www.xeno-canto.org), Cloudbirders (www.cloudbirders.com), Forest Type (<https://www.forestry.gov.my/index.php/en/2016-06-07-02-31-39/2016-06-07-02-35-17/forest-Type>)

Open Sciences

Open scientific publications, Open research data, Open source software

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Predicted distribution of the number of non-threatened hornbill species across Sarawak based on MaxEnt distributions of the black, bushycrested, oriental pied, rhinoceros, and wreathed hornbill.



17.1 BIODIVERSITY DATA CROWDSOURCING: A WAY FORWARD

Malaysia

Challenge

The main issues in biodiversity data inventory are fragmented monitoring efforts whereby many organizations act independently to develop their own databases and platforms, hindering comprehensive biodiversity data collection. Monitoring efforts of biodiversity data is still spatially and temporally fragmented, and taxonomically biased, which affects the long-term data availability. The absence of such a framework limits data collection to government agencies and university researchers, impeding the progress in biodiversity management in Malaysia. This restricts the ability to track changes in biodiversity over time, which is crucial for future conservation efforts.

Contribution

This paper focused on an examination of the existing biodiversity inventory system in Malaysia and the specific needs of a mobile crowdsourcing platform for the nation's forest stakeholders. Beyond merely assessing these requirements, the study also delves into the perspectives of forest stakeholder representatives regarding data sharing, the utility of crowdsourcing, and the role of citizen science. Crowdsourcing is positioned as a vital collaborative endeavor, bridging the gap between forest stakeholders and the broader community. The research has effectively mapped out a strategic pathway for data crowdsourcing in Malaysia, illuminating a progressive direction for future initiatives.

Citation

Azmy, S. N., Tarmidi, Z., Hassan, N., Johari, S. A., & Shamsir Omar, M. S. (2024). Biodiversity data crowdsourcing: a way forward. *IOP Conference Series: Earth and Environmental Science*.1412(1), 12023. <http://doi.org/10.1088/1755-1315/1412/1/012023>.

Data Uses

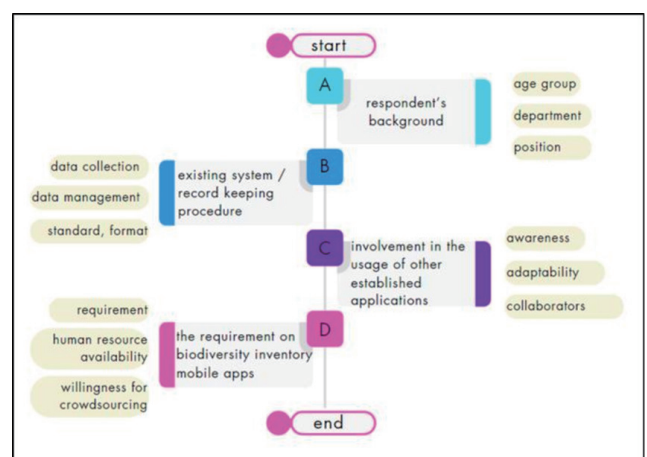
GBIF, Taxonomic Databases Working Group (TDWG), Darwin Core (DwC), MyGDI standard, iNaturalist, MyBIS app, and iSpot app, conducts surveys among respondents through online questionnaires and structured interviews.

Open Sciences

Open scientific publications, Open research data, Citizen science

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4. Faculty of Tropical Forestry, Universiti Malaysia Sabah, Malaysia



The sections addressed in data collection.



17.2 DATA SYNTHESIS FOR BIODIVERSITY SCIENCE: A DATABASE ON PLANT DIVERSITY OF THE INDIAN HIMALAYAN REGION

India

Challenge

Biodiversity data is the basis for supporting evidence-based conservation strategies that in turn help mitigate the impacts of climate change, meeting the multiple targets of UN sustainable development goals (SDGs). But the vast majority of these databases still lack adequate geographic coverage, particularly from the biodiversity hotspot regions of developing countries, thus severely limiting their utility and generalizability.

Contribution

A comprehensive and curated plant database of the Indian Himalayan Region (IHR) has been developed. Assembling biodiversity data with endemic and threat status holds promise to guide taxa and area-specific conservation efforts at a regional scale (e.g., across the IHR), with potential for interlinkages with global conservation efforts.

Citation

Wani, S. A., Khuroo, A. A., Zaffar, N., Rafiqi, S., Farooq, I., Afzal, S., & Rashid, I. (2024). Data synthesis for biodiversity science: a database on plant diversity of the Indian Himalayan Region. *Biodiversity and Conservation*, 33(12), 3377-3397. <https://doi.org/10.1007/s10531-024-02784-2>

Data Uses

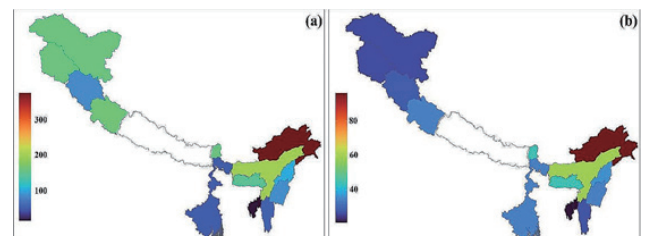
Botanical Survey of India (BSI), eFlora of India, India Biodiversity Portal, and Flowers of India, India Biodiversity Portal, Environmental Information System, Plants of the World Online (POWO)

Open Sciences

Open scientific publications; Open research data

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Choropleth maps representing the number of a) endemic and b) threatened plant species in each state of the Indian Himalayan Region. The colour scale indicates the number of species.

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Note: The abbreviations in open science: OSP: Open scientific publication; ORD: Open research data; OSS: Open source software; OER: Open educational resources; CS: Citizen science.

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